

PHARMACOLOGY OF THE CENTRAL NERVOUS SYSTEM

CENTRAL NERVOUS SYSTEM

- The functional unit of the central nervous system (CNS) is the neuron
- neuron is the **primary site of action** for most CNS Drugs
- neurons are capable of transmitting and receiving information to and from:-
 - ✓ other neurons
 - ✓ peripheral end organs, such as muscle cells, glandular cells, and specialized receptors, for example, those involved with proprioception, temperature sensing, and so on

CENTRAL NERVOUS SYSTEM...

- Like the peripheral nervous system, CNS involves the release of neurotransmitters that diffuse across the synaptic space to bind to specific receptors on the postsynaptic neuron
- Difference of CNS from peripheral nervous system
 - ❖ much more **complex circuitry** for the CNS and far greater number of synapses in the CNS
 - ❖ The CNS contains **powerful networks** of **inhibitory** neurons that modulate CNS
 - ❖ CNS communicates through the use of more than **10** different neurotransmitters.
 - » only two primary neurotransmitters, acetylcholine and NE for PNS.

CENTRAL NERVOUS SYSTEM...

- ➡ In CNS, receptors at synapses are coupled to ion channels
 - Binding of NT to postsynaptic membrane results in rapid but transient opening of ion channels.
- ↪ Change in ion composition across the membrane of the neuron
 - Produce depolarization or hyperpolarization
 - EPSP & IPSP

CNS pathways

➤ Excitatory Pathways

- Stimulation of excitatory neuron causes movement of ions that result in a **depolarization** of the postsynaptic potentials (**EPSP**) by influx of cations (Na^+).

➤ Inhibitory Pathways

- Stimulation of inhibitory neurons cause movement of ions that results in **hyper-polarization** of synaptic membrane (**IPSP**) by influx of chloride (Cl^-) or efflux of (K^+)

CNS Neurotransmitters

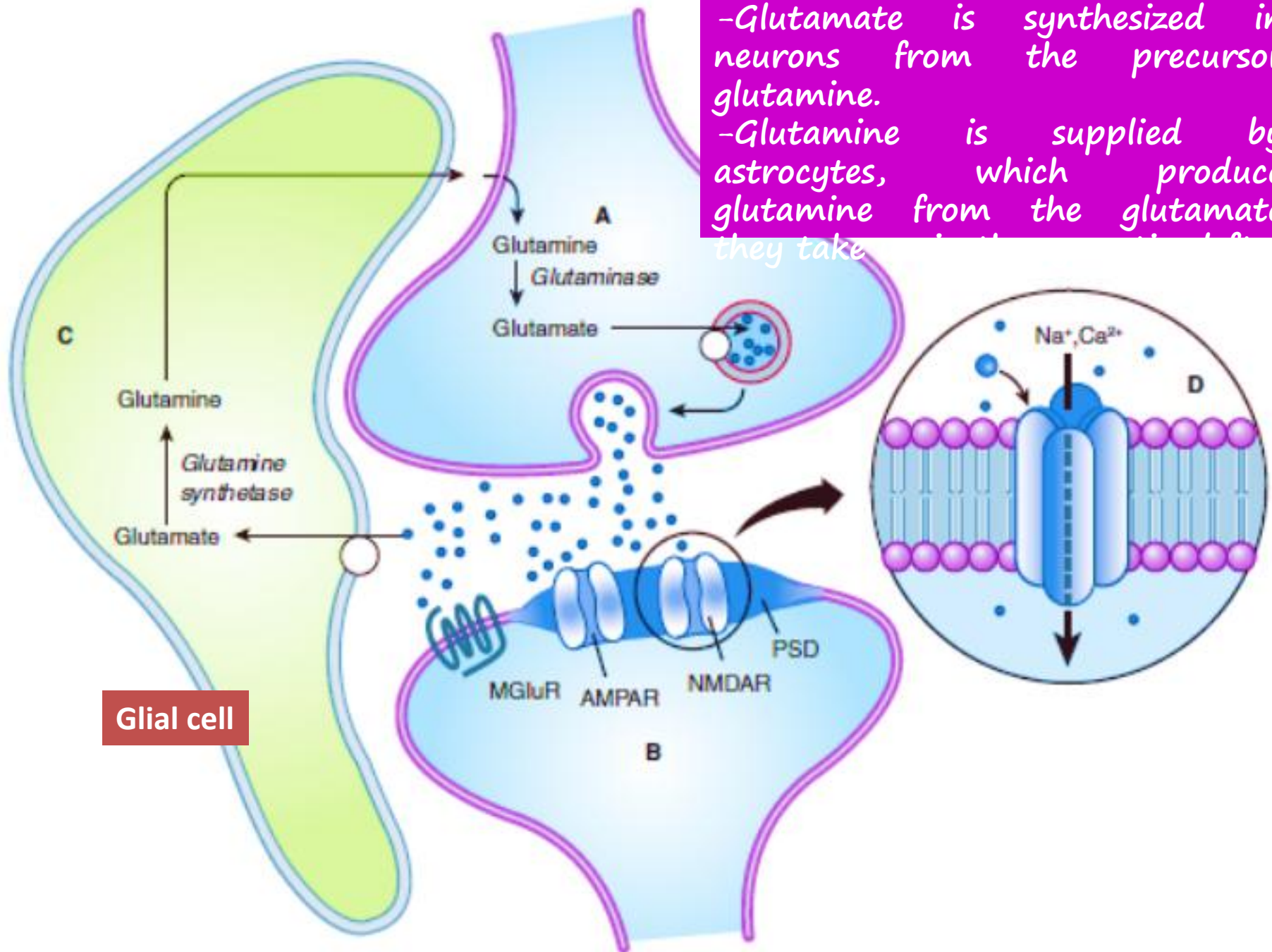
Glutamate & Aspartate: Excitatory amino acids (EAAs)

➔ Glutamate:

- The most common excitatory neurotransmitter in the CNS
- Cause influx of cations (+charge) into the postsynaptic neurons
- Binds to
 - **Ionotropic** glutamate receptors (**NMDA** receptors (N-methyl-D-aspartate), **AMPA** receptors (α -amino-3-hydroxyl-5-methyl-4-isoxazolepropionate), and **kainate** receptors).
 - **Metabotropic** glutamate receptors (mGluRs)

-Glutamate is synthesized in neurons from the precursor glutamine.

-Glutamine is supplied by astrocytes, which produce glutamine from the glutamate they take



CNS Neurotransmitters...

Glycine & γ -Amino-Butric-Acid (GABA)

- inhibitory transmitters primarily in the spinal cord & in the brain respectively
- Aspartate & Glycine form an excitatory/inhibitory pair in the spinal cord comparable to the excitatory/ inhibitory pair formed by Glutamate & GABA in the brain
- Glycine binds to an ionotropic receptor, which allows for Cl⁻ influx
- Strychnine is a glycine antagonist which can bind to the glycine receptor without opening the chloride ion-channel

CNS Neurotransmitters...

GABA can bind to

A. Ionotropic GABA receptors (GABAA and GABAC)

- induce Cl^- influx when activated
- Cl^- -influx leads to an accumulation of negative charge, which moves the membrane potential further away from reaching threshold (i.e., the neuron is “inhibited”).

B. Metabotropic GABA receptor (GABAB)

- activates K^+ channels and blocks Ca^{2+} channels
- resulting in a net loss of positive charge
- which leads to hyper polarization of the postsynaptic cell

CNS Neurotransmitters...

- **Acetylcholine (ACh)** is a neurotransmitter used in
 - PNS, NMJ & CNS (fore brain)
 - Role of cholinergic pathways
 - Arousal
 - learning
 - Cognitive functions- short-term memory
 - In disease conditions (Alzheimer's disease & Parkinsonism)

CNS Neurotransmitters...

Biogenic amines:

- ♠ Biogenic amines are a group of neurotransmitters with an amine group in their structure.
- ♠ They comprise the catecholamines Dopamine, Norepinephrine, and Epinephrine, as well as Histamine and Serotonin

Dopamine:

- ♠ Dopamine is involved in many forebrain circuits
 - ✓ Associated with **Emotion**, **Motivation**, and **Reward**
- ♠ It acts on G protein-coupled receptors, and its action can be either via D1 receptors or via D2 receptors.

CNS Neurotransmitters...

Dopamine Effects:

♠ Implicated in

- Motor control (parkinsonism is due to dopamine deficiency)
- Behavioral effects (excessive dopamine activity is implicated in schizophrenia)
- Hormone release (↓prolactin & GH release)
- Nausea and vomiting in chemoreceptor trigger zone (CTZ)

CNS Neurotransmitters...

➤ Norepinephrine & Epinephrine

- Adrenergic receptors (α and β) and their subtypes have been described in the CNS
 - all are GPCRs
 - Wakefulness and alertness
 - Control of mood (functional deficiency resulting depression)
- Epinephrine is found only in very low concentrations in the mammalian CNS
 - Unlikely to play a major role as a neurotransmitter.

CNS Neurotransmitters...

➤ 5-Hydroxytryptamine (Serotonin)

- Can have both excitatory and inhibitory effects
- It is involved in a multitude of pathways that regulate
 - ✓ mood,
 - ✓ emotion,
 - ✓ several homeostatic pathways
- Most serotonin receptors are metabotropic
- There is only one ionotropic receptor (5HT3), which is permeable to cations, therefore, excitatory

CNS Neurotransmitters...

5-HT....

➡ The 5-HT receptors fall into four broad classes:

① 5-HT₁ receptor

- Involved in **temperature regulation** and associated with **mood** and **anxiety**.
- 5-HT₁ receptor agonists, such as **buspirone** are used to treat anxiety and panic disorders.
- Sumatriptan activate **5-HT_{1D}** receptors (for headache)

CNS Neurotransmitters...

② 5-HT₂ receptor

- Target of various hallucinogenic drugs

③ 5-HT₃-receptor

- Involved in vomiting and emesis
- 5-HT₃ antagonists (e.g. ondansetron) -management of vomiting.

④ 5-HT₄, 5-HT₅, 5-HT₆, and 5-HT₇ ?

Summary of some neurotransmitters in the CNS

Neurotransmitter		Postsynaptic effect
Amino acids	Acetylcholine (ACh)	Excitatory
	Glutamate	Excitatory
	γ -Aminobutyric acid (GABA)	Inhibitory
	Glycine	Inhibitory
Biogenic amines	Dopamine	Excitatory (via D1 receptors)
		Inhibitory (via D2 receptors)
	Norepinephrine	Excitatory
	Epinephrine	Excitatory
	Serotonin	Excitatory or Inhibitory
	Histamine	Excitatory
Purines	Adenosine triphosphate (ATP)	Excitatory/neuromodulatory
Neuropeptides	Substance P	Excitatory
	Met-enkephalin	Inhibitory
	Opioids	Inhibitory
	Adrenocorticotropin	Excitatory

Ergotrophic- favor organisms capacity to expend energy

Trophatrophic-promote rest & reconstitution of energy stores

NORADRENALINE

SEROTONIN

**Ergotrophic
Vigilance**

**Trophatrophic
Impulse**

Anxiety
Irritability

Cognitive Function
Mood
Emotion

Motivation

Appetite
Sex
Aggression

**Pleasure
Drive**

DOPAMINE

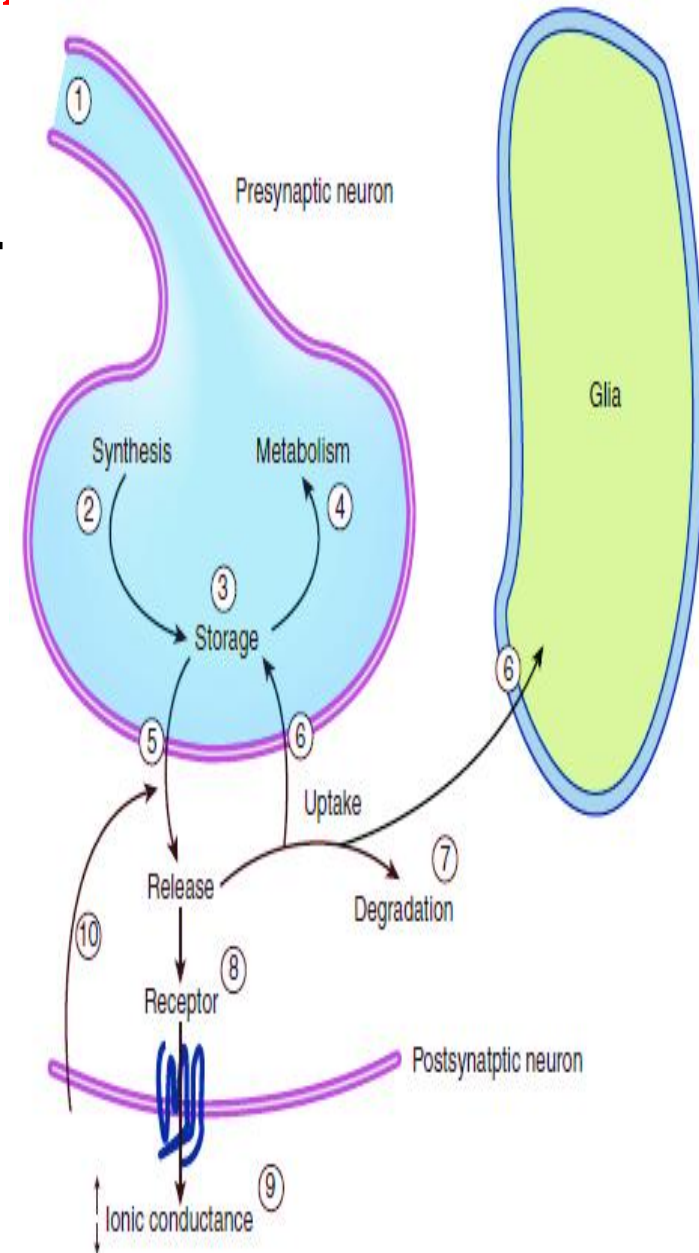
Neurotransmitter	Postsynaptic effect
Acetylcholine	involved in arousal, short term memory, learning and movement
Norepinephrine	involved in arousal, wakefulness, mood and cardiovascular regulation
Dopamine	involved in emotion, reward systems, motor control , hormone release and stimulate CTZ
Serotonin	feeding behavior, control of body temperature, modulation of sensory pathways including nociception (pain sensation), regulation of mood and emotion, and sleep/wakefulness, control vomiting

<i>Neurotransmitter</i>	<i>Postsynaptic effect</i>
GABA	increases Cl ⁻ influx into the postsynaptic neuron, resulting in hyperpolarization , mediates the majority of inhibitory postsynaptic potentials
Glycine	increases Cl ⁻ influx into the postsynaptic neuron, resulting in hyperpolarization
Glutamate	mediates excitatory Na ⁺ influx in to the postsynaptic neuron
Substance p	mediates nociception(pain) within the spinal cord
Met-enkephalin	mediates analgesia as well as other central nervous systemic effects

Sites of drug action which can alter synaptic transmission

Drugs can alter synaptic transmission

- ① Action potential in presynaptic fiber
- ② Synthesis of transmitter
- ③ Storage
- ④ Metabolism at nerve ending
- ⑤ Release
- ⑥ Reuptake into the nerve ending or
✓ uptake into a glial cell
- ⑦ Degradation at synapse
- ⑧ Receptor for the transmitter
- ⑨ Receptor-induced increase or
✓ decrease in ionic conductance
- ⑩ Retrograde signaling



Central Nervous System Drugs

- The central nervous system (CNS) drugs act on the brain and spinal cord
 - ✓ used widely for medical and nonmedical purposes.
- **Medical applications include treatment of**
 - ✓ psychiatric disorders, suppression of seizures, relief of pain, and production of anesthesia.
- **nonmedical applications for their**
 - ✓ stimulant, depressant, euphoriant, and other “mind-altering” **abilities.**

■ Sedatives-hypnotics and Anxiolytics Drugs

Introduction

- Anxiety and insomnia are common complaints
 - ✓ we can treat them by using the same drugs with **different dosages**(higher dose for insomnia)
- **Anxiety:** is an unpleasant state of tension, apprehension, or uneasiness a fear that seems to arise from a sometimes unknown source.
- **Insomnia:** refers to difficulty of falling asleep or staying asleep or having non-refreshing sleep.

■ Sedatives-hypnotics and Anxiolytics Drugs...

- **Sedative:** A drug that subdues excitement and calms the subject without inducing sleep, though drowsiness may be produced.
- Sedation refers to decreased responsiveness to any level of stimulation; is associated with some **decrease in motor activity and ideation**.
- **Hypnotic:** A drug that induces and/or maintains sleep, **similar** to normal arousable sleep.

Antianxiety drugs

Pathophysiology

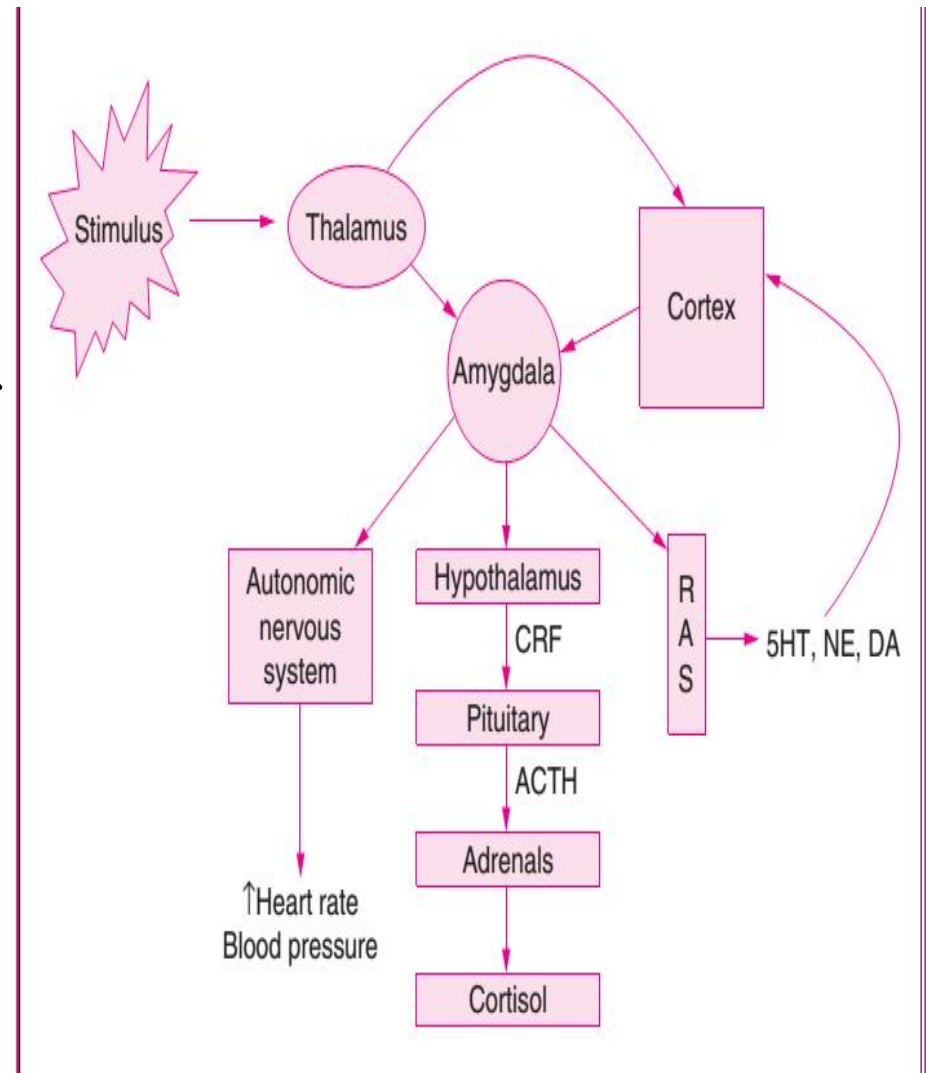
- thalamus and amygdala are important in the generation of a **normal fear response** and play a central role in most anxiety disorders.
- The thalamus passes information to higher cortical centers for finer processing and to the amygdala for rapid assessment of highly charged emotional information.
- The amygdala serves to provide emotional valence or the emotional importance of the information

Pathophysiology...

- Direct and indirect connections to the reticular activating system (RAS), a region spanning the medulla, pons, and midbrain, help to regulate **arousal, vigilance, and fear**.
- 5HT and NE, which have their primary origins in the RAS.
- Amygdala sends projections to the hypothalamus influencing the ANS to affect **HR, BP, and stress-associated changes**.
- The amygdala also influences the **HPA axis**, leading to a cascade of stress hormones.
- One such hormone is cortisol, which, if elevated for prolonged periods, can have damaging effects on the brain and other organ

neurotransmitters involved in mediating anxiety disorders

- Disorders involving anxiety are the most common mental disturbances.
- ✓ The physical symptoms of severe anxiety are similar to those of fear such as:-
 - ✓ tachycardia,
 - ✓ sweating,
 - ✓ trembling,
 - ✓ palpitations



- A. Drugs used for anxiety called “**antianxiety agents**” or “**anxiolytics**”
- B. A *sedative* drug- decreases activity, moderates excitement
- ✓ usually given during daytime hours
 - ✓ although they may make the patient drowsy, they usually do not produce sleep
 - ✓ produces **drowsiness** and **facilitates** the onset and maintenance of a state of sleep that resembles natural sleep
 - ✓ allows the patient to fall asleep and stay asleep
 - ✓ are given at night or hour of sleep

- Antianxiety drugs

- are used primarily to treat anxiety disorders

- The three main types of antianxiety drugs are:

- Benzodiazepines

- barbiturates and

- Buspirone

❖ Buspirone

- Its structure and mechanism of action differ from those of other antianxiety drugs
 - ✓ doesn't affect GABA receptors like the benzodiazepines
- interact with a number of neurotransmitter systems in the brain
- it appears that its **clinically relevant effects** are mediated through interactions at the serotonin(5-HT_{1A}) receptor, where it acts as a **partial agonist**.
- can relieve some symptoms of anxiety in doses that do **not cause sedation**

❖ **selective anxiolytic effects of Buspirone**

- ✓ relieves anxiety without causing **marked sedative, hypnotic, or euphoric** effects
- ✓ has no **anticonvulsant** or **muscle relaxant** properties
- ✓ not effective in blocking the acute withdrawal syndrome resulting from abrupt cessation of use of benzodiazepines or other sedative-hypnotics

■ ***Pharmacokinetics***

- ✓ It is absorbed rapidly, more than 95% bound to plasma proteins, undergoes extensive first-pass effect, and is metabolized in the liver .
- ✓ It is eliminated in urine and stool

❖ **Buspirone has several advantages, including:**

- ✓ less sedation,
- ✓ no increase in CNS depressant effects when taken with alcohol or sedative–hypnotics,
- ✓ lower abuse potential,
- ✓ no rebound anxiety or withdrawal signs on abrupt discontinuance

■ *Adverse effects*

- Like the benzodiazepines, buspirone appears to be safe even when given in very high doses
 - ✓ The most common adverse reactions to buspirone include:
 - ✓ Dizziness, light –headedness, insomnia, rapid heart rate, palpitations, headache
 - ✓ In therapeutic doses, buspirone has little or no sedative effect
 - ✓ **Abuse, dependence, and withdrawal** have not been reported
 - ✓ does not produce any cross-tolerance to the benzodiazepines

■ Clinical indication

❖ to treat generalized anxiety states

- ✓ *is as effective as the benzodiazepines in the treatment of general anxiety*
- ✓ However, the full anxiolytic effect of buspirone takes **several weeks** to develop, whereas the anxiolytic effect of the benzodiazepines is maximal after a **few days** of therapy
- ✓ Because of its slow onset of action, buspirone is **ineffective** when quick relief from anxiety is needed.(not useful for panic attack)

■ **Drug interactions**

- ✓ Unlike other antianxiety drugs, buspirone doesn't interact with alcohol or other CNS depressants.
- ✓ When buspirone is given with MAOIs, hypertensive reactions may occur

- The three main classes of synthetic drugs used as sedatives and hypnotics are:-

1. Benzodiazepines

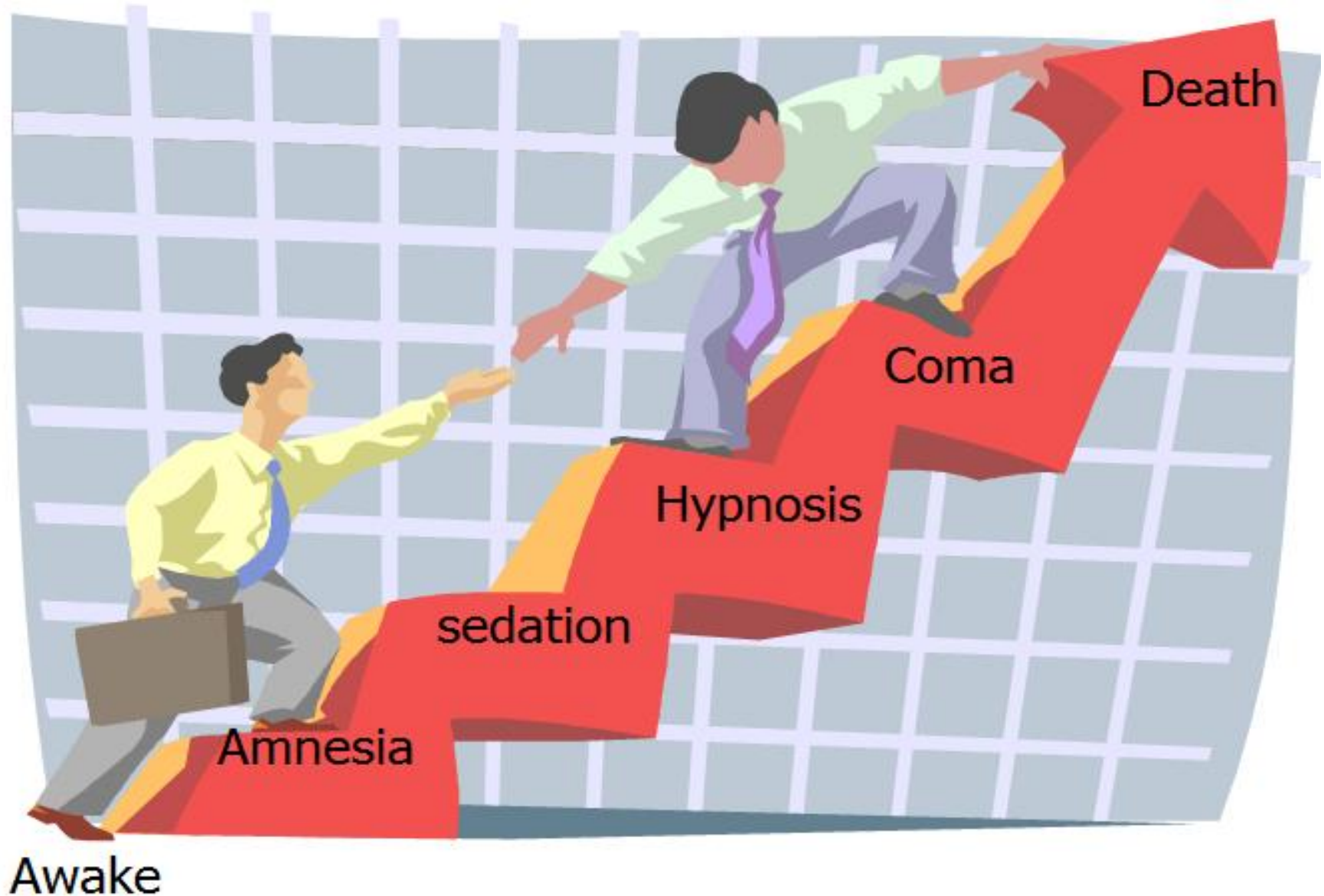
2. Barbiturates (were used before benzodiazepines)

3. Non-benzodiazepine-non-barbiturate drugs

➤ Zolpidem, chloral hydrate, paraldehyde etc

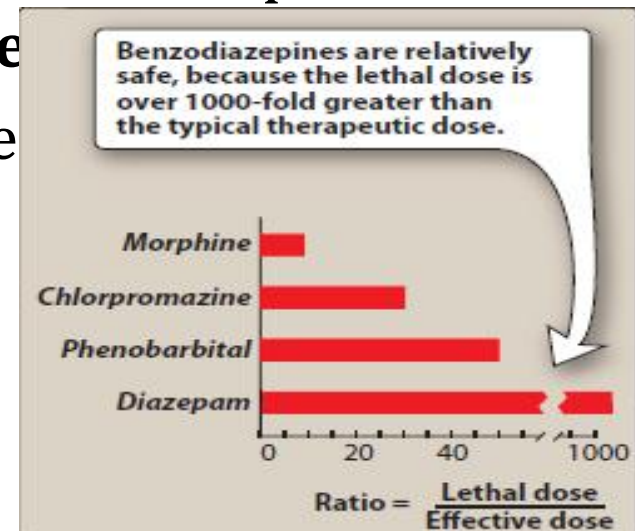
- Other sedatives may include sedating **antidepressants**, **antihistamines**, **antipsychotics**

Sedative-hypnotics



■ Benzodiazepines

- Are the most commonly used group of **anxiolytics** and **sedative-hypnotics**
 - Most of these drugs possess **anxiolytic**, **sedative-hypnotic**, and **anticonvulsant** properties.
 - largely replaced barbiturates and *meprobamate* in the treatment of anxiety and insomnia, b/c benzodiazepines considered to be **safer** and more **effective**
 - Commonly not necessarily the best choice
 - ✓ Anxiety or insomnia.
 - Certain antidepressants, such as
 - ✓ SSRI----- preferable for Anxiety
 - Non-benzodiazepine hypnotics
 - antihistamines
- may preferable for insomnia.



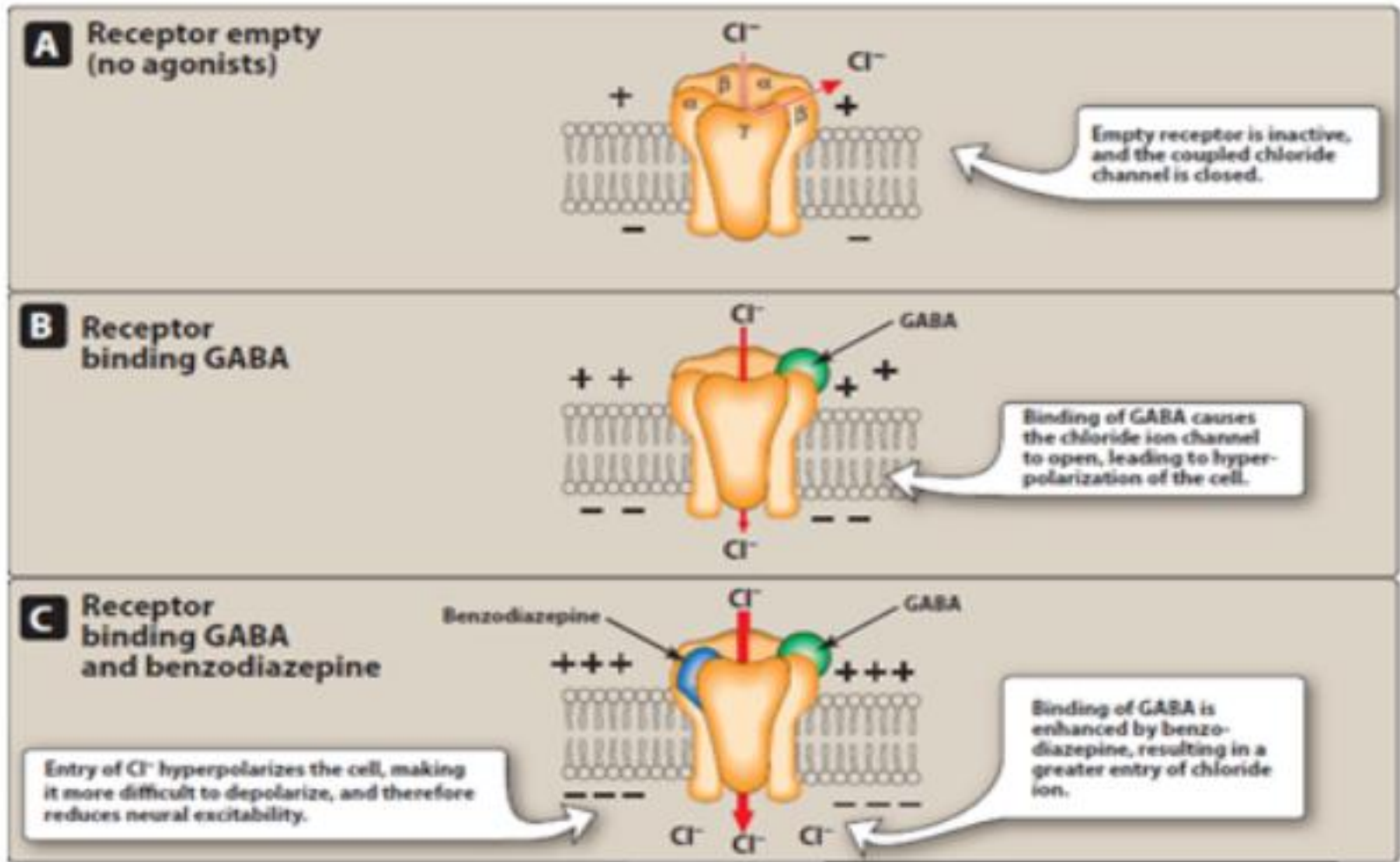
Benzodiazepines...

- Benzodiazepines used primarily for their sedative or hypnotic effects include
 - ✓ Diazepam, Flurazepam ,Nitrazepam,
Alprazolam ,Temazepam, Triazolam , Estazolam, quazepam,
- Benzodiazepines used primarily for the treatment of anxiety include:
 - ✓ Diazepam, Chlordiazepoxide, Oxazepam, Lorazepam,
Alprazolam, clonazepam, clorazepate,
- used for as Anticonvulsant
 - ✓ Diazepam, Lorazepam, Clonazepam ,Clobazam

▪ **MOA**

- bind to benzodiazepines site on neuronal GABA receptor , **enhancing the effect of GABA** and allowing more chloride ions to flow into the post-synaptic neuron.
 - This depresses the nerve impulses, causing them **to slow down or stop.**
- thereby inhibiting neuronal circuits in the limbic system of the brain
- Act on **GABAA receptors**
- Benzodiazepines are unable to produce effect in the **absence** of GABA

MOA



By Abebew Tegegne

- **low dosages of benzodiazepines:-**

- ✓ decrease anxiety by acting on the limbic system and other areas of the brain that help regulate emotional activity

- **higher dosages of benzodiazepines:-**

- ✓ induce sleep, probably because they depress the RAS (reticular activating system) of the brain which controls arousal and sleep-wake transitions

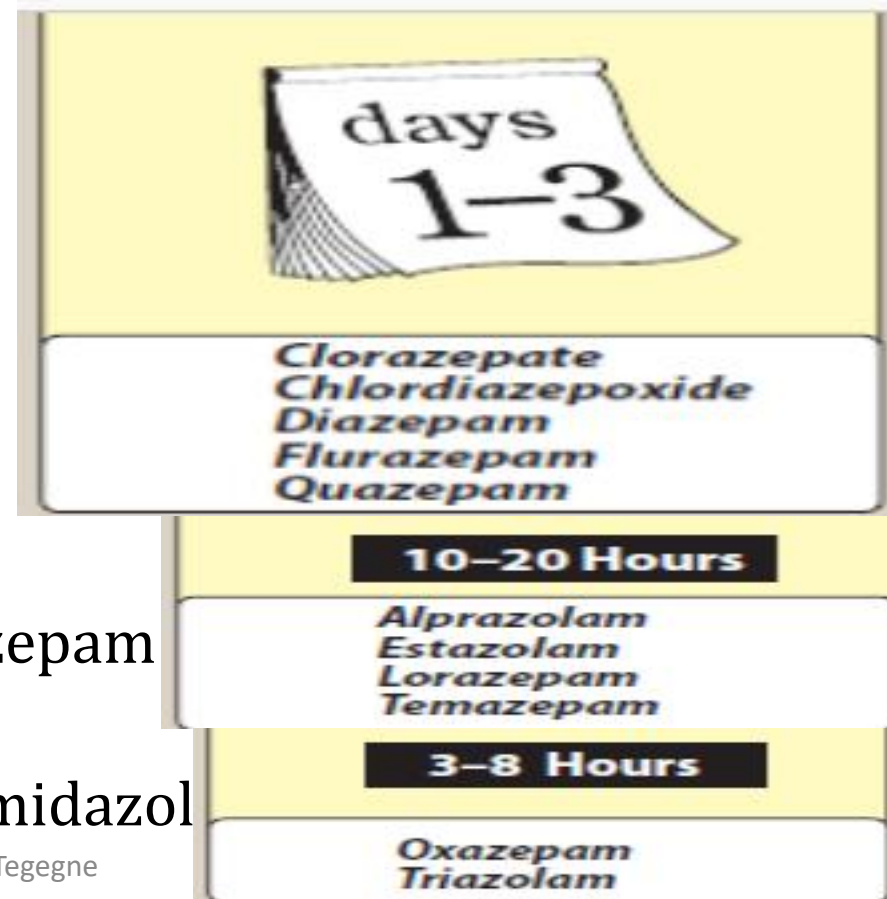
- Benzodiazepines increase **total sleep time** and reduce the **number of awakenings**

- duration of rapid -eye -movement (REM) sleep
 - ✓ REM sleep is associated with dreaming
 - ✓ Benzodiazepines reduce the **amount of time spent** in the deep sleep stages (before REM)
 - ✓ This decrease in deep sleep is accompanied by a reduction in **nightmares**

■ Pharmacokinetics

- Primary **difference** between various benzodiazepines is in pharmacokinetics
- Benzodiazepines are absorbed rapidly and completely from the GI tract , distributed widely in the body and Rapidly penetrate the brain
- Some benzodiazepines such as **diazepam** and **lorazepam** may also be given IV
- All benzodiazepines are metabolized in the liver and excreted primarily in urine

- All benzodiazepines cross the **placenta** and may depress the CNS of the newborn if given before birth.
- The benzodiazepines are **not recommended** for use during pregnancy.
- Nursing infants may also, be exposed to the drugs in breast milk
- **Long acting** – eg
 - ✓ diazepam, chlordiazepoxide,
 - ✓ flurazepam, clonazepam
 - Chlordiazepoxide---longest
- **intermediate acting-** eg temazepam
- **Shorter acting** –eg triazolam, midazolam



❖ Therapeutic uses

- small differences in their relative anxiolytic, anticonvulsant, and sedative properties.
 - ✓ choosing benzodiazepine considers **duration of action & pharmacokinetic**

1. Anxiety disorders:

- ✓ Benzodiazepines are effective for the treatment of the anxiety symptoms
- ✓ also useful in treating anxiety related to depression and schizophrenia.
- ✓ These drugs should be **reserved** for severe anxiety only and not used to manage the stress of everyday life.
 - b/c of their **addiction potential**, they should only be used for short periods of time

2.Treating insomnia

- Choice of specific benzodiazepam depends on pharmacokinetic criteria
 - Tend to decrease the **latency to sleep onset** and **increase non-rapid eye movement sleep**
- **Temazepam:** useful in patients who experience frequent wakening and given 1 to 2 hours before bedtime.---**slow onset**
- **Triazolam:** useful in patients who have difficulty in **going to sleep**
 - ✓ Tolerance is fast and withdrawal of the drug often results in rebound insomnia. is **not a preferred agent**, and it is best used **intermittently**
- In general, hypnotics should be given for only a limited time, usually less than **2 to 4 weeks**.
- Long-acting ***flurazepam*** is rarely used, due to its extended half-life,

3. Amnesia:

- The shorter-acting agents are often employed as **premedication** for anxiety-provoking and unpleasant procedures, such as endoscopy, dental procedures...
 - ✓ **Midazolam** is used to facilitate amnesia while causing sedation prior to anesthesia.

4. Treating alcohol withdrawal symptoms

- ✓ Withdrawal from long-term high-dose use of alcohol or sedative-hypnotic drugs can be life threatening if physical dependence is present.
- ✓ Benzodiazepines, such as chlordiazepoxide and diazepam are sometimes used to **lessen the intensity** of these withdrawal symptoms

5. Treating anxiety and seizure disorders

- *Clonazepam* is occasionally used as an adjunctive therapy for certain types of seizures,
 - ✓ whereas *lorazepam* and *diazepam* are the drugs of choice in terminating **status epilepticus**

6. producing skeletal muscle relaxation.

- Diazepam is useful in the treatment of skeletal muscle spasms, such as
 - ✓ muscle strain, and in treating spasticity from degenerative disorders, such as
 - **multiple sclerosis**
 - **cerebral palsy.**

❖ Dependence

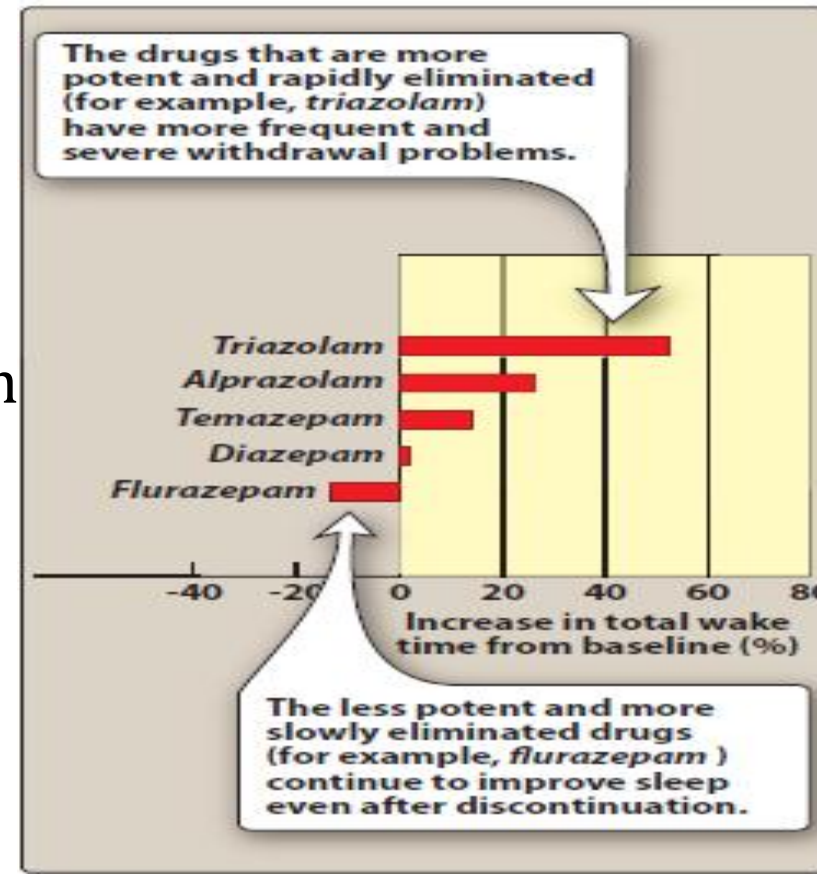
- Psychological and physical dependence on benzodiazepines can develop if given **over high doses** and a **prolonged period**.
- Abrupt discontinuation of the benzodiazepines results in **withdrawal** symptoms,
- Benzodiazepines with a short elimination half-life, such as *triazolam*, induce more abrupt and severe withdrawal reactions than those seen with drugs that are slowly eliminated such as *flurazepam*

■ Adverse effects

❖ Benzodiazepines may cause:

- ✓ Amnesia, fatigue, muscle weakness, mouth dryness, nausea , and vomiting, dizziness, ataxia (impaired ability to coordinate movement).
- ✓ Unintentional daytime sedation, hangover effect (residual drowsiness on awakening), and rebound insomnia may also occur.
- ✓ have a potential for abuse, tolerance, and physical dependence High dose chronic therapy may lead to dependence
- ✓ antianxiety effects of the benzodiazepines are less subject to tolerance than the sedative and hypnotic effects

- Cross-tolerance exists among this group of agents and with ethanol.
 - ✓ Abrupt withdrawal may cause syndrome that mimics alcohol withdrawal (convulsion, hyperthermia, delirium)
- used cautiously in patients with
 - ✓ impaired liver or kidney function
 - ✓ elderly patients



■ **Contraindications**

- ✓ in patients with known hypersensitivity, psychoses, acute narrow-angle glaucoma, and shock
 - ✓ Also in patients in a coma or with acute alcoholic intoxication
 - ✓ pregnancy and lactation
-
- When benzodiazepines are taken with other CNS depressants (including alcohol and anticonvulsants), the result is enhanced sedative and CNS depressant effects including death

▪ **Benzodiazepine Antagonist**

➤ **Flumazenil**

- is a GABA-receptor antagonist that can rapidly reverse the effects of benzodiazepines.
- available for IV use
 - ✓ Onset is rapid but duration is short, with a half-life of about 1 hour
 - ✓ Frequent administration may be necessary to maintain reversal of a long-acting benzodiazepine.
 - ✓ Dizziness, nausea, vomiting, and agitation are the most common side effects

Therapeutic Disadvantages

Therapeutic Advantages

Benzodiazepines

- The benzodiazepines may disturb intellectual functioning and motor dexterity.
- The benzodiazepines have the potential for dependence, and withdrawal seizures may occur.

- Withdrawal of drug often results in rebound insomnia.

Clonazepam

Clorazepate

Chlordiazepoxide

Diazepam

Flurazepam

Quazepam

Alprazolam

Lorazepam

Temazepam

Triazolam

- Potential use in chronic therapy for seizures.

- These less potent and more slowly eliminated drugs show no rebound insomnia on discontinuation of treatment.

- Agent of choice in treating panic disorders.

Do not require Phase I metabolism and, therefore, show fewer drug interactions and are safer in patients with hepatic impairment.

■ Barbiturates

- ***Introduction:-*** were formerly the mainstay of treatment to sedate the patient or to induce and maintain **sleep**.
- have been largely replaced by the benzodiazepines primarily
 - 👉 because barbiturates induce tolerance, drug-metabolizing enzymes, physical dependence, and are associated with very severe withdrawal symptoms, less therapeutic index
 - ✓ hypnotic doses affect respiration & CVS functions
 - 👉 great effect on REM phase of sleep, hence there is hangover
 - ✓ ability to cause coma in toxic doses
- No antidote for toxicity

❖ The major pharmacologic action of barbiturates is to reduce overall CNS alertness

- Barbiturates used primarily as sedatives and hypnotics include:
 - ✓ Amobarbital, butabarbital, mephobarbital, pentobarbital, **phenobarbital**, secobarbital.
- **Low doses:-**depress the sensory and motor cortex in the brain, causing drowsiness.
- **High doses :-**may cause **respiratory depression** and **death** because of their ability to depress all levels of the CNS

- At low doses, the barbiturates produce sedation (calming effect, reducing excitement)
- At higher doses, the drugs cause hypnosis, followed by anesthesia (loss of feeling or sensation), and finally, coma and death

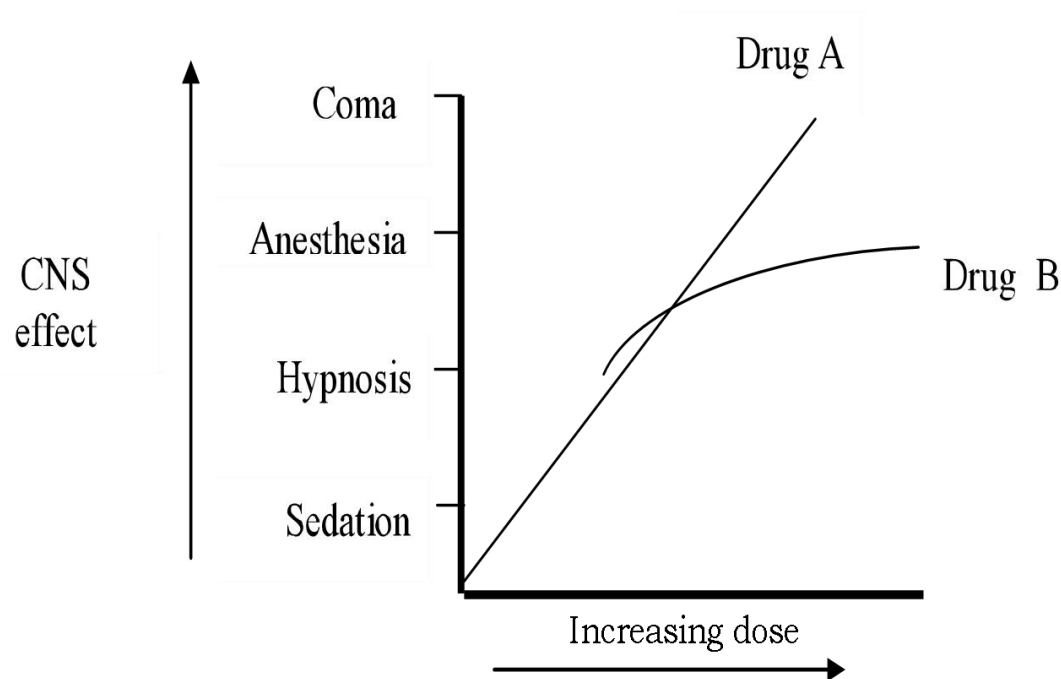


Fig. Dose - response curve for two hypothetical sedative - hypnotics.

■ **MOA**

- action of the barbiturates is due to their interaction with GABAA receptors, which enhances **GABAergic transmission**.
- The binding site is **distinct** from that of the benzodiazepines
- potentiate GABA action on **chloride entry** into the neuron by prolonging the duration of the chloride channel openings
 - In addition, barbiturates can block excitatory glutamate receptors
 - Anesthetic concentrations of pentobarbital also block high-frequency **sodium channels**
 - All of these molecular actions lead to decreased neuronal activity

- ***Pharmacokinetics***

- well absorbed from the GI tract, distributed rapidly, metabolized by the liver, and excreted in urine.
 - also given rectally, IM or IV
- Barbiturates vary markedly in lipid solubility and plasma protein binding

❖ Classified based on duration:-

- long acting(10-12hrs):

- ✓ barbital, phenobarbital, mephobarbital

- intermediate acting (6-8hrs):-

- ✓ aprobarbital, amobarbital, butabarbital

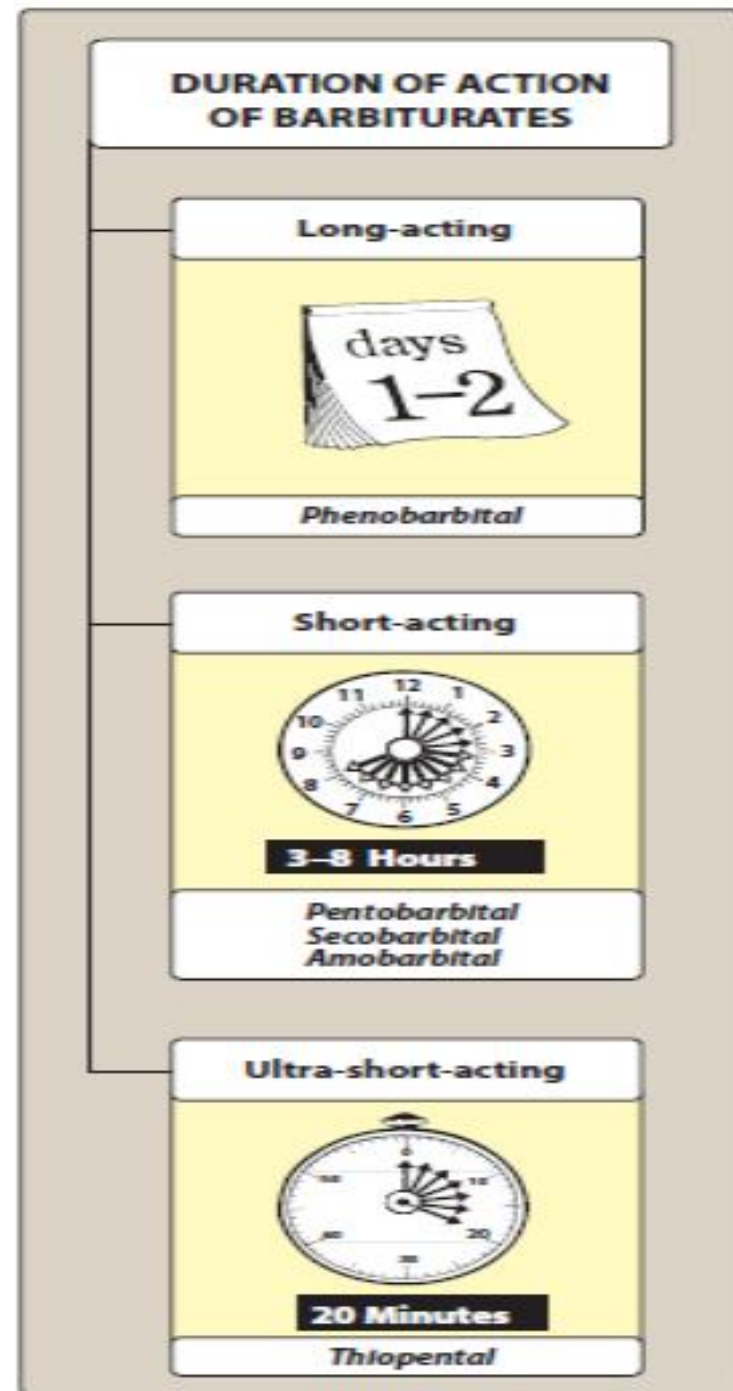
- short acting (3-6 hrs) :-

- ✓ pentobarbital, secobarbital

- ultrashortacting (<3hr):-

- ✓ thiopental, methohexital, hexobarbital:

- ✓ are used as anesthetics



■ **clinical indications**

- ❖ **Anesthesia:** *thiopental*, used iv to induce anesthesia but have largely been replaced by other agents
- ❖ **Anticonvulsant:** *Phenobarbital* is used in long-term management of tonic-clonic & refractory status epilepticus seizures.
 - ✓ due to **depress cognitive**, used only alternative is failed
- ❖ **Sedative/hypnotic:** used as mild sedatives to relieve anxiety, nervous tension, and insomnia----**suppress REM sleep**
 - ✓ no longer used accepted,
- *Butalbital* is used in combination products (with *acetaminophen* and *caffeine* or *aspirin* and *caffeine*) as a sedative to assist in the management of tension-type or migraine headaches.

- Patients develop tolerance to barbiturates more quickly than to benzodiazepines
 - physical dependence on barbiturates may occur even with a small daily dosage.
- ❖ In comparison, benzodiazepines are relatively effective and safe
- for these reasons, benzodiazepines have replaced barbiturates as the sedatives and hypnotics of choice.

■ Drug interactions

- Barbiturates may interact with many other drugs:
 - may cause excessive CNS depression with other CNS depressants (especially alcohol).
 - may reduce the effects of beta-adrenergic blockers (metoprolol, propranolol), corticosteroids, doxycycline, oral anticoagulants, hormonal contraceptives, quinidine, tricyclic antidepressants (TCAs) etc.
 - ✓ by increasing their metabolism
 - phenytoin, reduce the metabolism of phenobarbital, resulting in increased toxic effects.
 - Monoamine oxidase inhibitors (MAOIs) inhibit the metabolism of barbiturates, increasing their sedative effects

■ Adverse effects

- Drowsiness, lethargy, headache, depression
- mild bradycardia, hypotension
- Hypoventilation, spasm of the larynx (voice box) and bronchi, reduced rate of breathing, severe respiratory depression
- Vertigo, nausea and vomiting, diarrhea, epigastric pain, allergic reactions.

👉 Acute renal failure-oliguria & anuria

■ Adverse effects...

- Tolerance, physical and psychological dependence are likely
- Withdrawal reaction may be severe enough to cause death
- should be used with great caution during lactation
- should be administered with extreme caution to patients with
 - ✓ a history of drug abuse (eg, alcoholics and opiate abusers)
 - ✓ liver or kidney disease

Treatment of poisoning

- ✓ includes artificial respiration
- ✓ purging the stomach of its contents if the drug has been recently taken
- ✓ Hemodialysis may be necessary if large quantities have been taken
- ✓ Alkalinization of the urine often aids in their elimination
- ❖ *No specific barbiturate antagonist is available*

■ Non-benzodiazepines-non-barbiturates

❖ act as hypnotics for treatment of simple insomnia

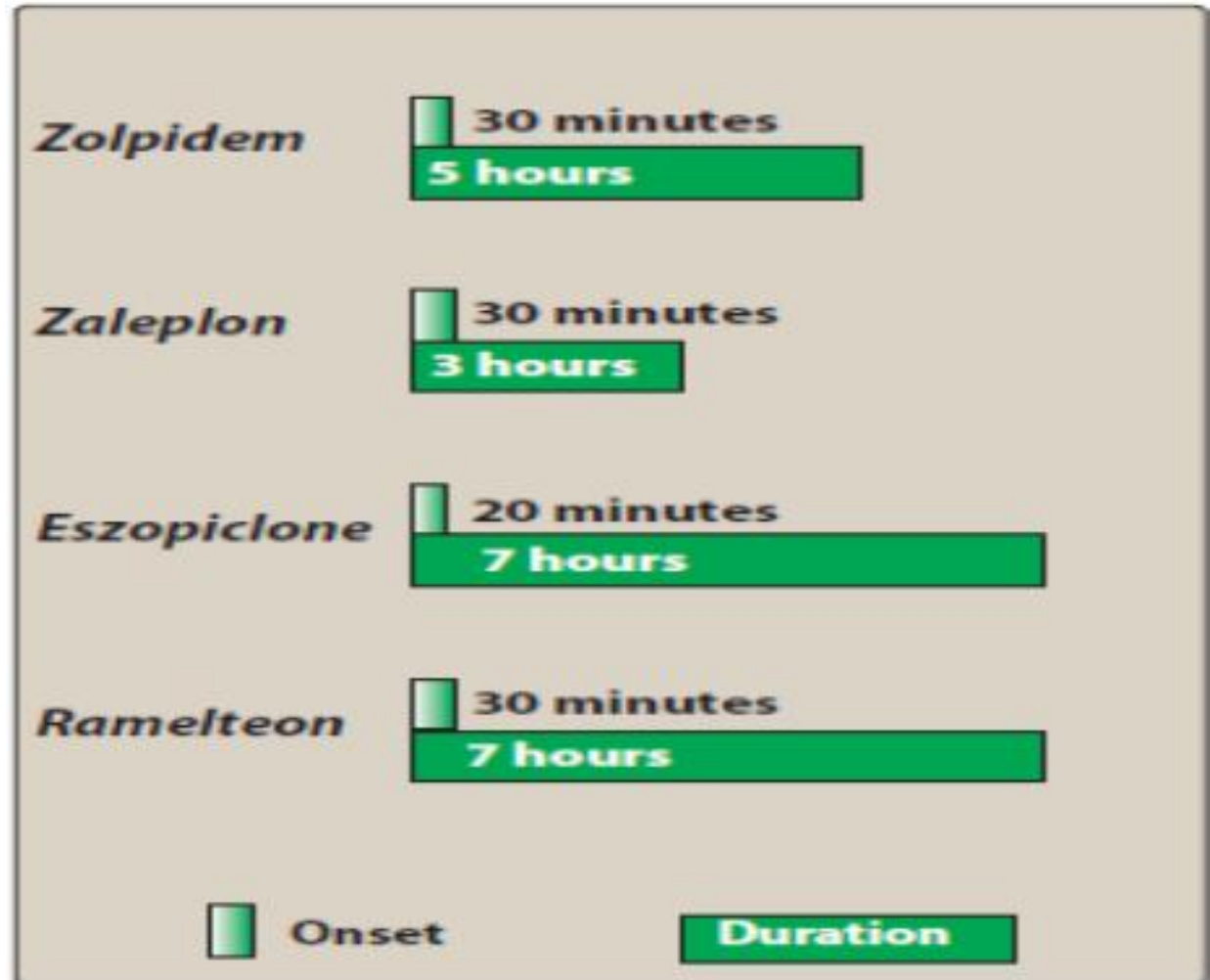
➤ Include:

✓ chloral hydrate,

✓ ramelteon,

✓ zaleplon,

✓ zolpidem



- **Zolpidem, zaleplon** and **Eszopiclone** are structurally unrelated to the benzodiazepines but bind to benzodiazepine receptors and facilitate GABA-mediated inhibition.
 - are absorbed rapidly from the GI tract, metabolized in the liver, and excreted in urine.
- Unlike the benzodiazepines, at usual hypnotic doses zolpidem and zaleplon do **not significantly alter** the various sleep stages and, hence, are often the **preferred hypnotics** .

- ***Zolpidem*** has no anticonvulsant or muscle-relaxing properties.
 - ✓ It shows few withdrawal effects, exhibits minimal rebound insomnia, and little tolerance occurs with prolonged use.
- *rifampin*, which induce liver enzyme, shorten the half-life of *zolpidem*, and drugs that inhibit the CYP3A4 isoenzyme may increase the half-life.
- Adverse effects of *zolpidem* include nightmares, agitation, anterograde amnesia, headache, GI upset, dizziness, and daytime drowsiness.
- All three agents are controlled substances.

❖ Ramelteon

- ✓ is a selective agonist at the MT1 and MT2 subtypes of melatonin receptors.
- ✓ Melatonin is a hormone secreted by the pineal gland that helps to maintain the circadian rhythm underlying the normal **sleep–wake cycle**.
- ✓ treatment of insomnia characterized by difficulty falling asleep
- ✓ It has minimal potential for abuse, and no evidence of dependence or withdrawal effects has been observed.
- ✓ adverse effects include dizziness, fatigue...
- ✓ *Ramelteon* may also increase prolactin levels.

■ Clinical use

- Non-benzodiazepines–non-barbiturates are typically used for:
 - ✓ treatment of simple insomnia
 - ✓ sedation before surgery and before EEG studies.

■ *Drug interactions*

- ❖ When used with other CNS depressants, additive CNS depression occurs, resulting in drowsiness, respiratory depression, stupor, coma, or death

■ *Adverse effects*

- ❖ The most common dose -related adverse reactions involving non-benzodiazepines-non-barbiturates include:
 - nausea and vomiting, gastric irritation, hangover effects (possibly leading to respiratory depression or even respiratory failure)

❖ **Chloral hydrate:-**

- is derivative of acetaldehyde
 - is converted to the active metabolite, **trichloroethanol**, in the body.
- is an effective sedative and hypnotic
- induces sleep in about **30 minutes** and the duration of sleep is about **6 hours**
- **Adverse effects**
- is irritating to the gastrointestinal tract and causes epigastric distress also produces an unusual, unpleasant taste sensation it synergizes with ethanol

❖ Antihistamines (H1 antagonists)

- e.g. diphenhydramine, promethazine, and hydroxyzine
- ✓ since they produce some degree of **sedation**, they have been used as sedative–hypnotics
- ✓ this sedation is usually considered a side effect of their antihistaminic activity
- ✓ but in some cases the sedation is sufficient to allow the drugs to be used in the treatment of anxiety and sleep disturbances
- ✓ they have undesirable side effects (such as **anticholinergic effects**) that make them less useful than the benzodiazepines and the non-benzodiazepines

❖ Antihistamines (H1 antagonists)...

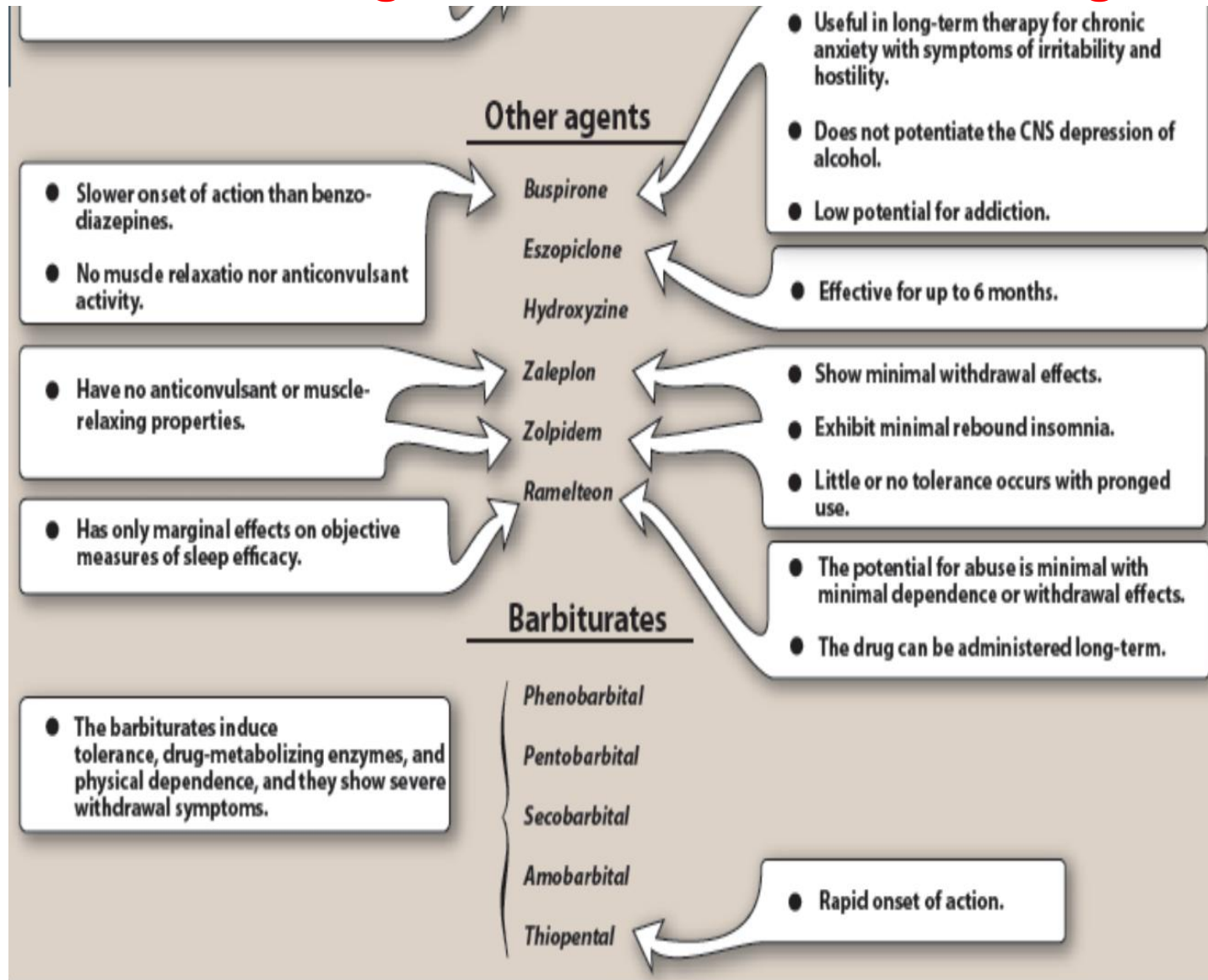
- Used for milder forms of situational insomnia
- Also as antianxiety (which is associated with anesthesia and surgery)
 - ✓ **Hydroxyzine hydrochloride** is the antihistamine with the greatest use in the treatment of anxiety.

❖ **β -Adrenoceptor Blocking Agents**

- **Also are useful in some forms of anxiety**
- ✓ particularly those that are characterized by **performance anxiety (stage fright)**
- ✓ can lessen and prevent the appearance of many of the autonomic responses associated with anxiety.
 - ✓ Eg. tremors, sweating, tachycardia, and palpitations.

Disadvantage

Advantage



❖ Antidepressants

- Eg: the tricyclic antidepressants and the selective serotonin reuptake inhibitors (SSRIs)
 - They are very important for the treatment of psychotic depression
- Also shown to be effective when used in the treatment of several anxiety disorders,
- including general anxiety, obsessive-compulsive disorder, and several phobias

Because the SSRIs are less toxic than the tricyclic antidepressants, their use in the treatment of anxiety is safer and less likely to produce serious side effects.

■ Ethanol

- Ethanol is CNS depressant, producing sedation and, ultimately, hypnosis with increasing dosage
- has anxiolytic, sedative and hypnotic effects
 - but its toxic potential outweighs its benefits
- readily absorbed orally and has a volume of distribution close to that of total body water.
- Ethanol is metabolized primarily in the liver, first to acetaldehyde by **alcohol dehydrogenase** and then to acetic acid by **aldehyde dehydrogenase**

- Ethanol synergizes with many other sedative agents such as benzodiazepines, antihistamines, or barbiturates
- Chronic consumption can lead to severe liver disease, gastritis, and nutritional deficiencies.
- Cardiomyopathy is also a consequence of heavy drinking
- **Treating alcohol withdrawal**
 - treatment of choice for alcohol withdrawal are the benzodiazepines.
 - Carbamazepine is effective in treating convulsive episodes during withdrawal.

drugs used in alcoholic people to stop drinking

- Disulfiram
- Naltrexone
- **Acamprosate**
- **Serotonergic Agents**
- **Anticonvulsants**

■ Disulfiram:

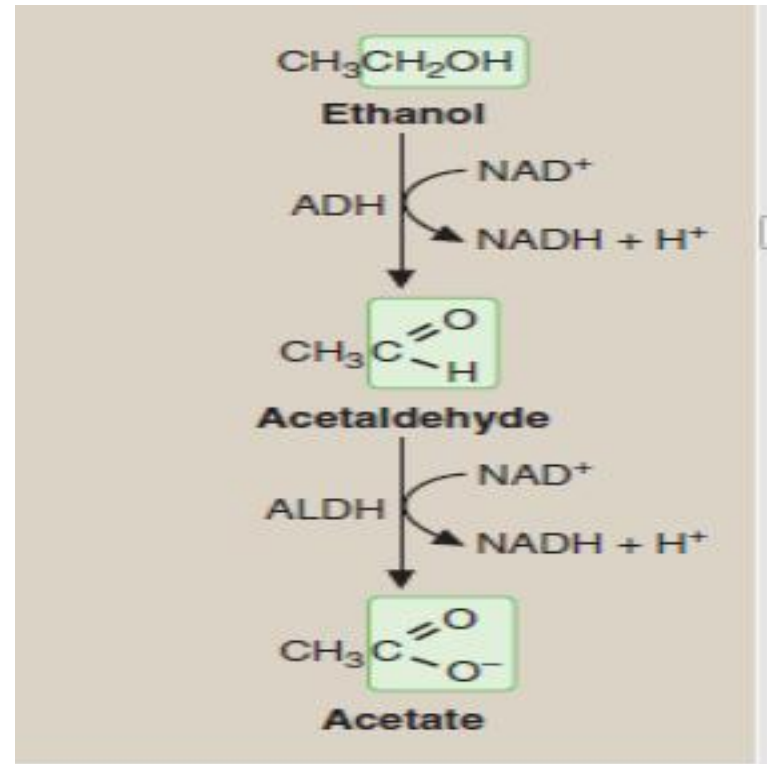
- inhibit acetaldehyde dehydrogenase
- blocks the oxidation of acetaldehyde to acetic acid
 - » results in the accumulation of acetaldehyde in the blood
 - » Causes flushing, tachycardia, hyperventilation, and nausea
- Drugs that cause disulfiram like effects- metronidazole, cefoperazone, cefotetan, griseofulvin

- More severe reactions could include myocardial infarction, congestive heart failure, respiratory depression, and death.

- **contraindicated**

- metronidazole
- ingested alcohol,
- psychosis,
- cardiovascular disease,
- severe pulmonary disease,
- chronic renal failure,
- diabetes, older than 60 years.
- peripheral neuropathy, seizures, or cirrhosis

Hepatotoxicity is a rare but potentially fatal adverse effect.



- **Naltrexone:**

- is a long-acting opiate antagonist
 - available orally or as a long-acting injectable
 - approved for the treatment of alcohol dependence and should be utilized in conjunction with supportive psychotherapy
 - is better tolerated than disulfiram and does not produce the aversive reaction that disulfiram does
- It is believed blockage of μ opioid receptors, which reduces the **reinforcing effects of alcohol leading to decreased feelings of intoxication and fewer cravings.**

■ **Methanol**

- Cause CNS depression in part through GABA mimetic effect
- Effects of methanol:- respiratory failure, ocular damage, metabolic acidosis

■ **Metabolism**

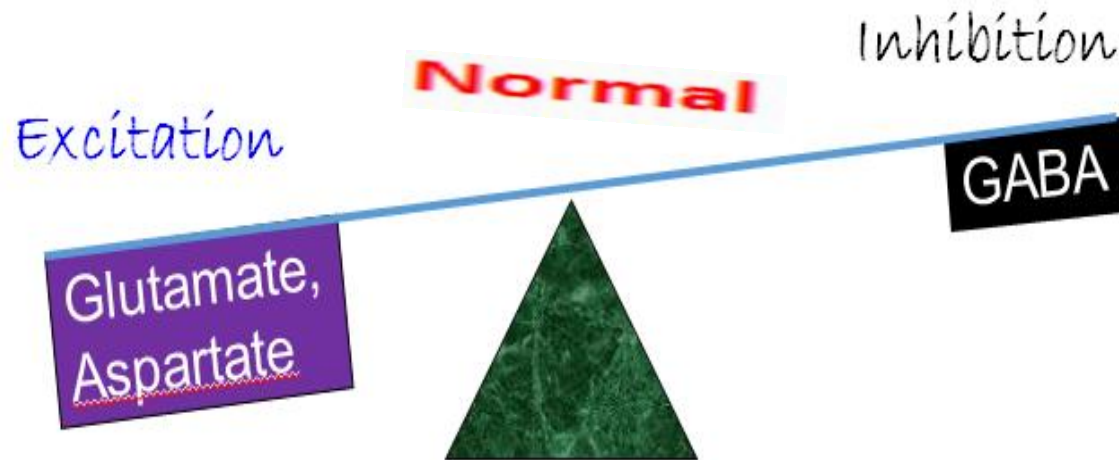
- Methanol is changed in to formaldehyde by the action of alcohol dehydrogenase
- Formaldehyde in to formic acid by aldehyde dehydrogenase

■ **Treatment of overdose**

- Fomepizole- long acting inhibitor of alcohol dehydrogenase

SEIZURE AND EPILEPSY

- **Seizure-** described as an abnormal disturbance (hyperexcitability) in the electrical activity in one or more areas of the brain



- ✓ **Epilepsy:** a clinical condition of recurrent (two or more) seizures due to a chronic, underlying process

⇒ **Characterized by recurrent seizures**

■ Etiology of seizures

👉 5–10% of the population will have at least one seizure

❖ **Acquired** (known cause)

– **A. Genetic seizures**

- These seizures result from an inherited abnormality in the CNS.

– **B. Structural/metabolic seizures**

- A number of causes, illicit drug use, tumor, head injury, hypoglycemia, meningeal infection, and the rapid withdrawal of alcohol from an alcoholic, high grad fever can precipitate seizures.

❖ **Idiopathic** (Unknown cause)

- ✓ When no specific anatomic cause for the seizure (70%)

Drugs as a cause of seizure

- **Drug withdrawal**
 - Alcohol
 - Barbiturates (short-acting)
 - Benzodiazepines (short-acting), Zolpidem
- **Drugs of abuse** (Amphetamine, Cocaine, Phencyclidine, Methylphenidate)
- Flumazenil
- Antidepressants (e.g., bupropion)
- Antipsychotics (e.g., clozapine)
- Lithium.....

■ **Principles of anticonvulsant therapy**

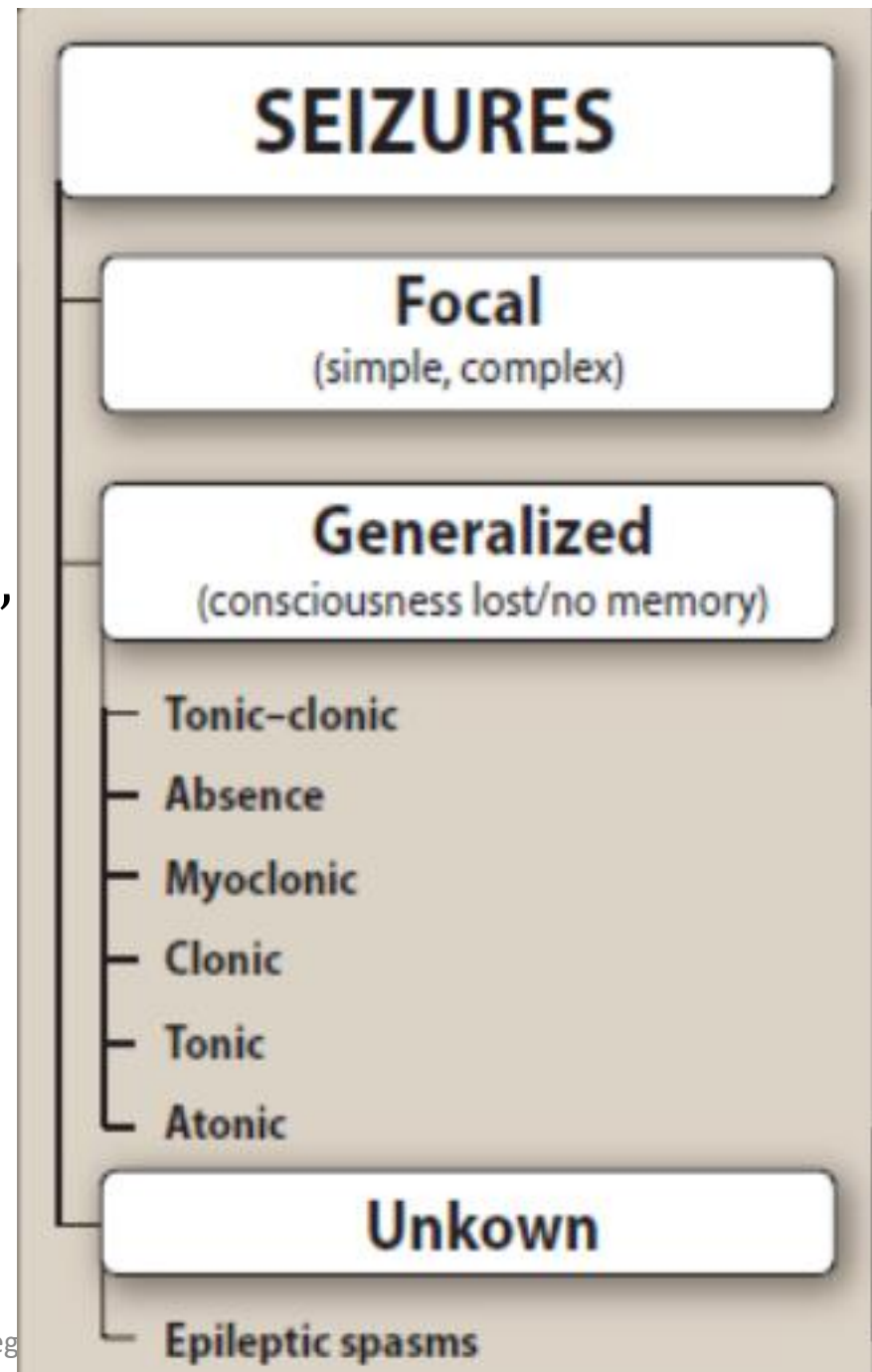
- Increase dose of suitable agent until desired effect is achieved or until toxicity prevents further increase
- Follow serum drug levels
- A second drug may be added if maximal doses of the initial drug fail.
- The initial drug should then be tapered and discontinued
- Abrupt discontinuance of an anticonvulsant may induce status epilepticus. (Always taper doses)

■ Principles of anticonvulsant therapy...

- ➡ Inform female patients of association with birth defects
- ➡ Anticonvulsants are additive with other CNS depressants
- ➡ Phenobarbital is considered safest during pregnancy
- therapeutic goal in epilepsy treatment is complete seizure control without excessive side effects
- overall, only about 40 to 60% of patients become totally seizure free with available drugs

■ Seizure classification

- categorized by:-
 - site of origin,
 - etiology,
 - electrophysiologic correlation,
 - clinical presentation .
- May be classified as
 - ✓ partial (focal) or generalized.
- **NB:** Patients often exhibit
 - ✓ more than one type



1. Partial or focal seizures

- arise from a localized area in the brain and cause specific symptoms.
 - often due to structural abnormalities such as scars, tumors or inflammation
- It can spread to the entire brain and cause a generalized
- **Further classified:-**
 - A. Simple partial seizures**
 - ✓ consciousness is retained
 - ✓ exhibits abnormal activity of a single limb or muscle
 - ✓ sensory distortions.
 - ✓ may occur at any age
 - ✓ Jerking movements of the face, arm, or leg

Partial or focal seizures ...

B. Partial complex seizure

- ✓ Begin in a focal area and spread
- ✓ Focal signs are often followed by loss of consciousness
- ✓ exhibit complex sensory hallucinations and mental distortion.
- ✓ Motor dysfunction may involve
- ✓ chewing movements, diarrhea, and/or urination.
- ✓ Simple partial seizure activity may spread to become complex and then spread to a secondarily generalized convulsion
- ✓ may occur at any age

2. Generalized seizures

- **Generalized seizures**

- ✓ may begin locally and then progress to both hemispheres
- ✓ Primary generalized seizures may be convulsive or nonconvulsive,
- ✓ immediate loss of consciousness.
 - include absence, myoclonic, and tonic-clonic types

➤ **Generalized tonic- clonic seizures/grand mal**

- loss of consciousness, followed by tonic (continuous contraction) and clonic (rapid contraction and relaxation) phases.
- may be followed period of confusion and exhaustion due to the depletion of glucose and energy stores
- Drugs of choice in Tonic-clonic
 - Valproic acid, Phenytoin, Carbamazepine

➤ **Myoclonic seizures**

- involve sudden, forceful contractions involving the musculature of the trunk, neck, and extremities.
- Treatment of choice
 - Valproic acid, clonazepam

➤ **Absence seizures/petit mal**

- ✓ a brief, abrupt, and self limiting loss of consciousness during which physical activity ,without muscle mov't
- Generalized seizures (in children or teens) which present as brief episodes of blank staring but no convulsions
- The patient stares and exhibits rapid eye-blinking, which lasts for 3 to 5 seconds.
- occur many times a day, and may go unnoticed by others.
- Drugs of choice in absence seizure
 - ethosuximide and Valproic acid

➤ Status epileptics

- an emergency situation characterized by continual seizure activity with no interruptions
 - two or more seizures occur without recovery of full consciousness
- may be focal or primary generalized, convulsive or nonconvulsive.
- may cause permanent brain damage, leading cause of death
- life threatening and requires emergency treatment usually consisting of administration of a fast-acting medication such as a benzodiazepine, followed by a slower-acting medication such as phenytoin.

- **Categorization of Anticonvulsants by Their Proposed Mechanism**

- **Type I**

- **enhancing sodium channel inactivation**

- Phenytoin, Carbamazepine, Oxcarbazepine, Lamotrigine, Felbamate

- **Type II**

- **Multiple actions: enhance GABAergic inhibition, reduce T-calcium currents, and possibly block sodium channels**

- Valproic acid, Benzodiazepines, Phenobarbital, Primidone

➤ **Type III**

- Block T-calcium currents only
 - Ethosuximide, Trimethadione

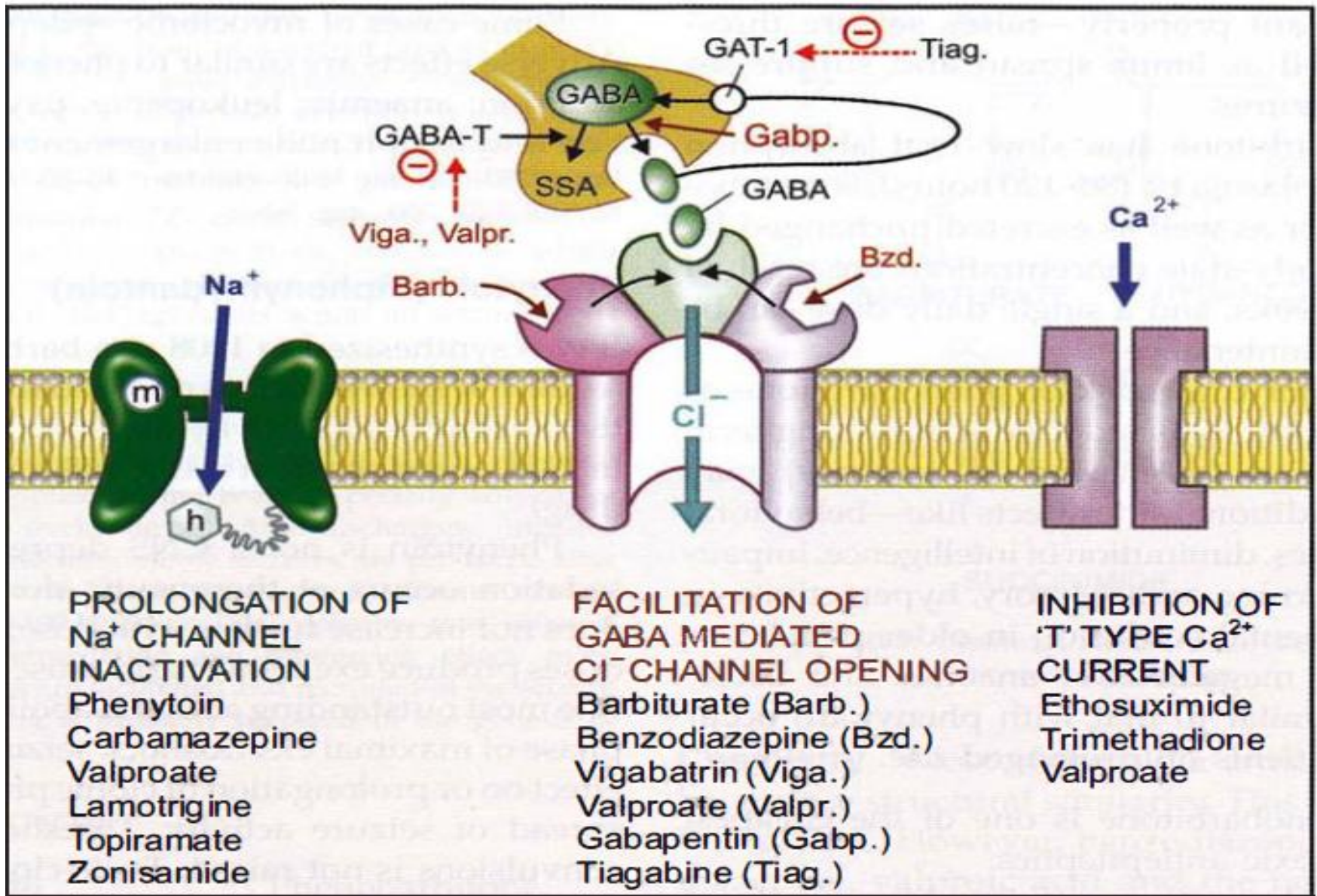
➤ **Type IV**

- Only enhances GABAergic inhibition
 - Vigabatrin

➤ **Non-categorized**

- Has no known effect on sodium channel, GABAergic inhibition, or T-calcium currents
- Gabapentin

Major mechanisms of anticonvulsant action



➡ phenobarbital

▪ *MOA*

- Enhances GABA mediated chloride influx
- At higher concentrations can block sodium channels

▪ *Pharmacokinetics*

- given IV/PO
- Slow onset (>30 min)
- Very long half life (>50 hrs)
- Metabolized by P450 enzymes in liver

- ***Indications-*** is one of the cheapest and least toxic antiepileptics.
- It has broad spectrum efficacy in **GTC, SP and CP seizures**
- less popular than carbamazepine, phenytoin or valproate.

■ ***Adverse effects***

- CNS depression, drowsiness
- may also cause nausea, vomiting, constipation, bradycardia, hypoventilation, skin rash, headache, fever, and diarrhea.
 - Some of these adverse effects may be reduced or eliminated as therapy continues.
 - Occasionally, a slight dosage reduction, without reducing the ability of the drug to control the seizures, will reduce or eliminate some of these adverse reactions.

➤ *decreased IQ in children treated chronically*

Drug interactions

- ✓ Stimulates P450 enzymes, additive with CNS depressants

- **Benzodiazepines**
 - Most benzodiazepines are **reserved** for emergency or acute seizure treatment due to **tolerance**.
 - *clonazepam* and *clobazam* as adjunctive therapy for particular types of seizures
- **Diazepam and lorazepam**

- ***MOA***

- Enhances GABA mediated chloride influx

- ***Pharmacokinetics***

- IV/PO/PR
 - Rapid onset(<5minutes)
 - Metabolized in liver
 - Has longer half life

■ **Clonazepam**

- ✓ very long acting with potent anticonvulsant activity.
- ✓ **sedation** and **tolerance** tend to limit its usefulness.
- ✓ **Drooling** and **hypersalivation** may be difficult in children and in infants.
- ✓ choice for emergency treatment of
 - status epilepticus,
 - serial seizures,
 - prolonged seizures
 - for prophylaxis of febrile seizures.
- ✓ IV is preferable for emergency treatment.

■ ***Drug interactions***

- Additive with CNS depressants
- Many drugs inhibit metabolism of diazepam

- **Phenytoin**

- barbiturate analogue,
- is not a CNS depressant; some sedation occurs at therapeutic doses

- ***MOA***

- Blocks sodium channels

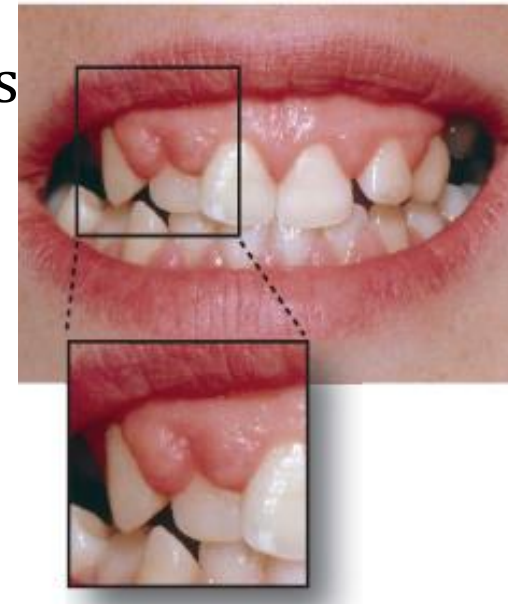
- ***Pharmacokinetics***

- ✓ PO/IV/IM, Oral absorption is slow but usually complete , is highly bound (about 90%) to plasma proteins, Slow onset, long half life, metabolized in liver.

❖ ***Indications*** -All types of seizure except **absence seizures**

▪ *Adverse effects*

- Nystagmus (involuntary eye mov't), ataxia, other CNS depressants, bone marrow suppression, **gingival hyperplasia**, hepatotoxicity, GI disturbances,
- IV administration may cause CNS depression, severe hypotension, arrhythmias, and hyperkinesia
- Teratogenicity- cleft lip and palate
- Osteomalacia (↓vitamin D)
- Megaloblastic anemia (↓folate)



Drug interactions

- Many drugs alter **metabolism** and **protein binding** of phenytoin and vice versa
- Drugs that compete for protein binding (eg, carbamazepine, sulfonamides, valproic acid)
 - Phenytoin levels in plasma are increased transiently
- Inhibitors of liver metabolism (Cimetidine, Isoniazid)
- Phenytoin induces hepatic drug metabolism (CYP2C and CYP3A families and the UGT enzyme)
 - Decreasing the effects of other antiepileptic drugs including carbamazepine, clonazepam, and lamotrigine₂₉

- **Interaction**

- isoniazid, chloramphenicol, sulfonamides, benzodiazepines, succinimides, and cimetidine all increase phenytoin blood levels
- barbiturates, rifampin, theophylline, and warfarin decrease phenytoin blood levels

- **CONTRAINDICATIONS**

- AV block, pregnancy

- **used cautiously**

- liver or kidney disease, neurologic disorders, hypotension, severe myocardial insufficiency

■ Carbamazepine

- a major drug in the treatment of seizure disorders.
- It has high efficacy, well tolerated by most patients, and fewer long-term side effects than other drugs.
 - Also has antidiuretic effect (increase ADH secretion)
 - Chemical structure similar to tricyclic antidepressants as high efficacy,
- **MOA**-Blocks sodium channels like phenytoin
- **Pharmacokinetics**-It is absorbed slowly and erratically (PO)
- Induces cytochrome P450, including its own metabolism (at high conc.)

- ***Indications***

- All types of seizure except absence seizures
 - the most effective drug for CPS and shares first choice drug status with phenytoin for GTCS and SP
 - But Carbamazepine has been shown to exacerbate or precipitate seizures in some patients, particularly those exhibiting generalized atypical absences.
- Bipolar depression, Mania
 - Trigeminal neuralgia (drug of choice)

- ***Adverse effects***

- **Drowsiness**, agranulocytosis or aplastic anemia (monitor blood counts), vertigo, nausea, vomiting
- Cause rashes and other allergic reactions
 - Teratogenic
 - little risk of cognitive impairment
 - Water retention and hyponatremia can occur in the elderly because it enhances ADH action

- **contraindicated** in patients with bone marrow depression or hepatic or renal impairment and during pregnancy

- ***Drug interactions***

- Do not administer to patients with MAOIs (potentiation may be lethal)

- Metabolism of **Carbamazepine** is induced by phenobarbitone, phenytoin, valproate and vice versa
- induce enzymes of haloperidol, oral contraceptives.
- A common interaction with erythromycin, fluoxetine, isoniazid inhibit metabolism of carbamazepine.
 - After a few days of antibiotic therapy, symptoms of carbamazepine toxicity develop;
- Cimetidine, propoxyphene, erythromycin and isoniazid also have been reported to inhibit metabolism of carbamazepine.
 - It is essential to monitor blood levels and adjust the dose

- **Valproic acid**
- marketed as both valproic acid and as sodium valproate
 - ***MOA***
 - Similar to phenytoin
 - Also inhibition of GABA transaminase
 - Blockade of T-type calcium channel
 - ***Pharmacokinetics***
 - PO/IV
 - well absorbed from the gastrointestinal tract and is highly bound (~90%) to plasma protein
 - Metabolized in liver
 - Inhibit P450 enzymes

■ *Indications*

- All seizure types
- particularly for combined disorders is highly effective
- Valproic acid is the drug of choice for absence seizures.
- It is an alternative/ adjuvant drug for GTCS, SPS and CPS.
- Myoclonic and atonic seizures---control is often incomplete, but valproate is the drug of choice.
- Mania and bipolar illness: as alternative to lithium.
- Valproate has some prophylactic efficacy in migraine.

■ *Adverse effects*

- ✓ Anorexia, vomiting, heart burn are common.
- ✓ Little sedation or other central effects-----at dose level
- ✓ **cognitive and behavioral effects are not prominent.**
- ✓ Rare/Severe/fatal hepatotoxicity (due to toxic metabolites), particularly below 3 yr
- ✓ Thrombocytopenia, hyperammonemia, **alopecia**, increased bleeding tendency
- ✓ Long-term use in young girls higher incidence of **polycystic ovarian** disease and menstrual irregularities.
- ✓ Contraindication with renal impairment or pregnancy-----

Teratogenicity

- **Interactions**

- ✓ increases plasma levels of phenobarbitone by inhibiting its metabolism.
- ✓ It displaces phenytoin from protein binding site and decreases its metabolism ---phenytoin toxicity.
- ✓ Valproate and carbamazepine induce each other's metabolism.
- ✓ Concurrent administration of clonazepam and valproate is contraindicated because **absence status** may be precipitated.
- ✓ Foetal abnormalities are more common if valproate and carbamazepine are given concurrently.

■ Ethosuximide

■ MOA

- Blocks T-calcium channels in thalamic neurons

■ Pharmacokinetics

- PO, Metabolized in liver, not protein bound

■ Indications

- **Only absence seizures**

■ Adverse effects

- For the most part, ethosuximide is a safe drug.
- Most of the side effects are dose related Headache, nausea, dizziness, vomiting, fatigue, ataxia, blurred vision, confusion, rashes, hepatotoxicity, blood dyscrasias, lupus like syndrome

Newer Antiepileptic Drugs (Second-Generation)

- 👉 Vigabatrin 1989
- 👉 Gabapentin 1993
- 👉 **Lamotrigine** 1994
- 👉 Topiramate 1996
- 👉 Tiagabine 1997
- 👉 levetiracetam 1999
- 👉 Oxcarbazepine 2000
- 👉 Zonisamide 2000

Newer agents differ from older drugs

- Relatively lack of drug-drug interaction (simple pharmacokinetic profile)
- Improved tolerability
- Cause fewer cognitive impairments than the older agents
- Topiramate may cause substantial cognitive impairment
- But they are costly with limited clinical experience

Lamotrigine

Pharmacological effects

- Block Na⁺ channels and glutamate receptors
- Also block voltage-dependent calcium channels
- effective in epilepsy syndromes with mixed, generalized seizure types
- Well absorbed from GIT
- Metabolized primarily by glucuronidation
- Does not induce or inhibit CYP-450 isozymes (its metabolism is inhibited by valproate)
- **Side effect:** Stevens-Johnson syndrome (skin rash), somnolence, blurred vision...

Gabapentin

- Structural analogue of GABA
- Act through calcium channel

Pharmacokinetics

- Not bound to proteins, BA decreases with increasing dose
- Not metabolized and excreted unchanged in urine, short $t_{1/2}$ (tid)
- Does not induce or inhibit hepatic enzymes (similar to lamotrigine)

Side effects: weight gain, somnolence, dizziness, ataxia, fatigue and nystagmus.

Uses: a second-line agent for patients with partial seizures or as an adjunct with other anti-epileptics

Topiramate

- Multiple mechanisms of action (Na, Ca & NMDA receptors)
- Effective for use in partial and primary generalized epilepsy
- It is also approved for prevention of migraine
- It inhibits CYP2C19 and is induced by phenytoin and carbamazepine.
- 70% of the drug is excreted unchanged by the kidneys

Adverse effects : cause cognitive disturbances, lethargy, and impaired mental concentration when given in large daily doses

- Nephrolithiasis occurs in 1.5% of patients, weight loss
- associated with acute narrow-angle glaucoma, word-finding difficulties, oligohidrosis, and metabolic acidosis

Felbamate

- has a broad spectrum of anticonvulsant action
 - Block voltage-dependent sodium channels
 - competing with the glycine coagonist binding site on the NMDA receptor
 - Blocking of calcium channels & potentiating GABA action
- It is an inhibitor of drugs metabolized by CYP2C19 and induces drugs metabolized by CYP3A4.
- It is reserved for use in refractory epilepsies because of the risk of aplastic anemia (about 1:3000) and hepatic failure.
- causes insomnia and irritability, headache, weight loss, and GI effects

Oxcarbazepine

- is a prodrug that is rapidly reduced to the 10-monohydroxy (MHD) metabolite responsible for its anticonvulsant activity.
- MHD blocks sodium channels, preventing the spread of the abnormal discharge.
- It is also thought to modulate calcium channels
- It is approved for use in adults and **children** with partial-onset seizures
- less potent inducer of CYP3A4 and UGT than carbamazepine.
- The adverse effect of hyponatremia limits its use in the elderly

Levetiracetam

- is approved for adjunct therapy of focal onset, myoclonic, and primary generalized tonic-clonic seizures in adults and children.
- The exact mechanism of anticonvulsant action is unknown.
- The drug is well absorbed orally and excreted in urine mostly unchanged (66%), resulting in few to no drug interactions.
- Dosage reductions are warranted for patients with renal impairment
- has a short half-life
- can cause mood alterations that may require a dose reduction or a change of medication.

Zonisamide

- a sulfonamide derivative that has a broad spectrum of action
- blockade of both voltage-gated Na channels & T-type Ca currents.
- Approved for use in patients with focal epilepsy
- The most commonly reported adverse events include ataxia, somnolence, agitation, and anorexia.
- May cause kidney stones (2.6% of pts)
- Oligohidrosis: patients should be monitored for increased body temperature and decreased sweating.
- contraindicated in patients with sulfonamide or carbonic anhydrase inhibitor hypersensitivity

- **Tiagabine**

- ***MOA*** -Blocks presynaptic GABA uptake
- ***Pharmacokinetics***-PO
- ***Indications***- primarily used in the treatment of partial complex seizures
- **Adverse effects** -Dizziness, nervousness, tremor
- **Drug interactions** -Cleared faster in patients taking other anticonvulsants
- Titrate dose over 6-20 weeks until effective Partial seizures

■ **Pregnancy and epilepsy**

- should prescribe only one drug at the lowest effective dosage to minimize teratogenic risks.
- Phenobarbital is the usual drug

■ **treatment of febrile seizures**

- Convulsions associated with fever often occur in children 3 months to 5 years of age.
 - Epilepsy later develops in 2 to 3% of children who exhibit one or more such febrile seizures.
- ✓ Phenobarbital is the usual drug,
 - ✓ diazepam is also effective.
 - ✓ Phenytoin and carbamazepine are **ineffective**,
 - ✓ valproic acid may cause hepatotoxicity in very young patients

- prophylactic treatment with anticonvulsant drugs only to patients at highest risk for development of epilepsy and for those who

- **STATUS EPILEPTICUS**

- Pharmacological treatment of choice at present consists of intravenous infusion of either diazepam or lorazepam or fosphenytoin.
- Fosphenytoin is a prodrug that is converted to phenytoin following parenteral administration.
- All of these drugs should be administered slowly to avoid respiratory depression and apnea.

Table 30.1: Choice of antiseizure drugs

<i>Type of seizure</i>	<i>First choice drugs</i>	<i>Second choice drugs</i>	<i>Alternative/Add-on drugs</i>
1. Generalised tonic-clonic/ simple partial with or without generalization	Carbamazepine, Phenytoin	Valproate, Phenobarbitone	Lamotrigine, Gabapentin, Topiramate, Primidone
2. Complex partial with or without generalization	Carbamazepine, Valproate, Phenytoin	Gabapentin, Lamotrigine	Clobazam, Zonisamide Topiramate, Tiagabine
3. Absence	Valproate	Ethosuximide, Lamotrigine	Clobazam, Clonazepam
4. Myoclonic	Valproate	Topiramate, Lamotrigine	Primidone, Clonazepam
5. Atonic	Valproate	Clonazepam, Clobazam	Lamotrigine
6. Febrile seizures	Diazepam (rectal)	—	—
7. Status epilepticus	Diazepam (i.v.) Lorazepam (i.v.)	Fosphenytoin (i.v.) Phenobarbitone (i.v., i.m.)	Gen. anaesthetics

Drugs for Parkinson disease

- disorder of muscle movement, characterized by:
 - tremors, muscular rigidity, bradykinesia (slowness in initiating and carrying out voluntary movements), and postural and gait abnormalities
- If untreated the symptoms progress over several years to end-stage disease
 - ✓ rigid,
 - ✓ unable to move,
 - ✓ unable to breathe properly
 - ✓ chest infections/embolism.
- **cause of Parkinson's disease is unknown for most patients**
- mostly affecting older people

- It is correlated with destruction of **dopaminergic** neurons in the substantia nigra
 - reduction of dopamine actions in the brain's basal ganglia system that are involved in **motor control**
- imbalance between the **excitatory cholinergic neurons** and the greatly diminished number of inhibitory **dopaminergic neurons**
 - Therapy is **aimed** at **restoring** dopamine in the basal ganglia and **antagonizing** the excitatory effect of cholinergic neurons

- **Classification of drugs**

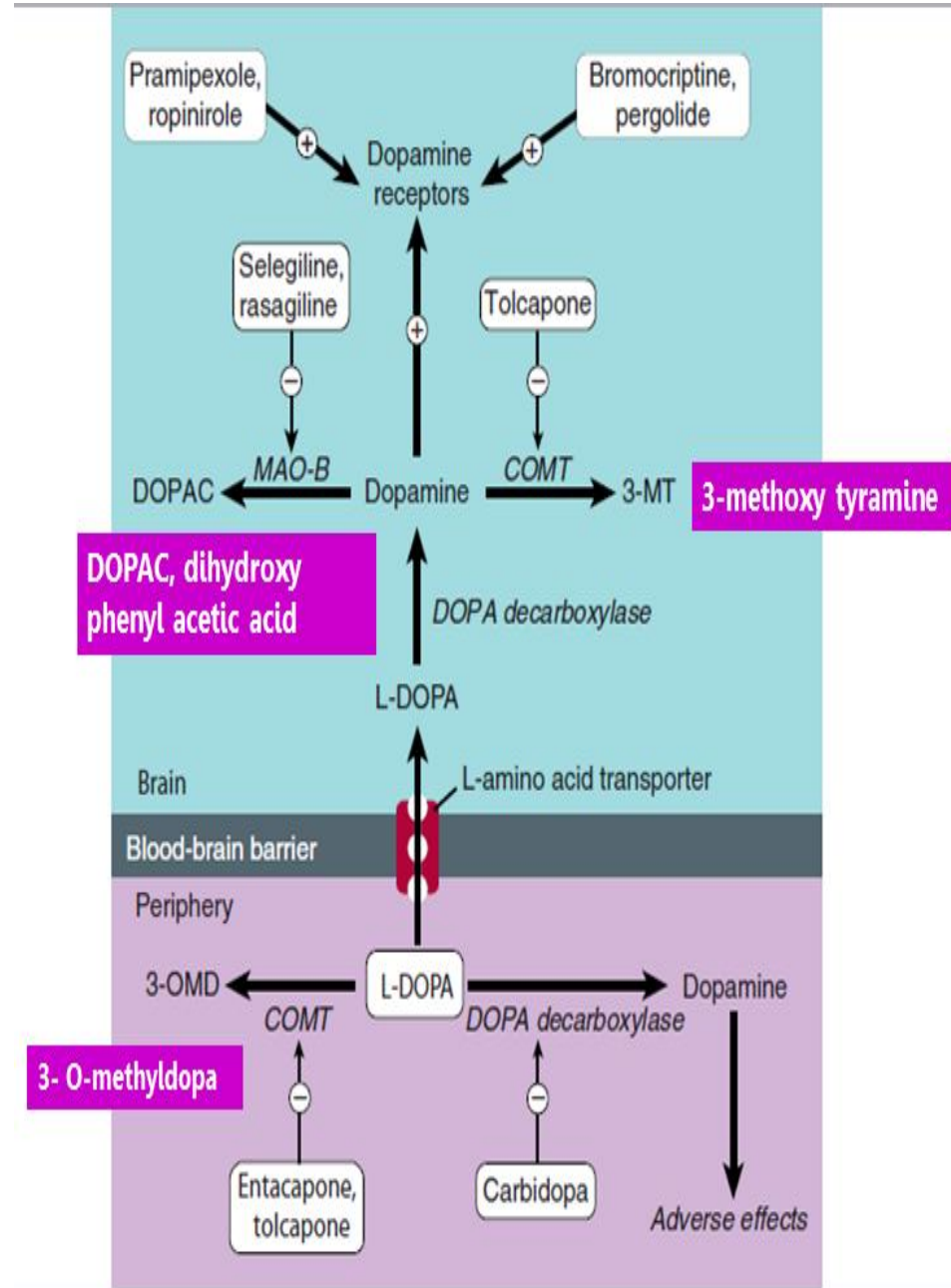
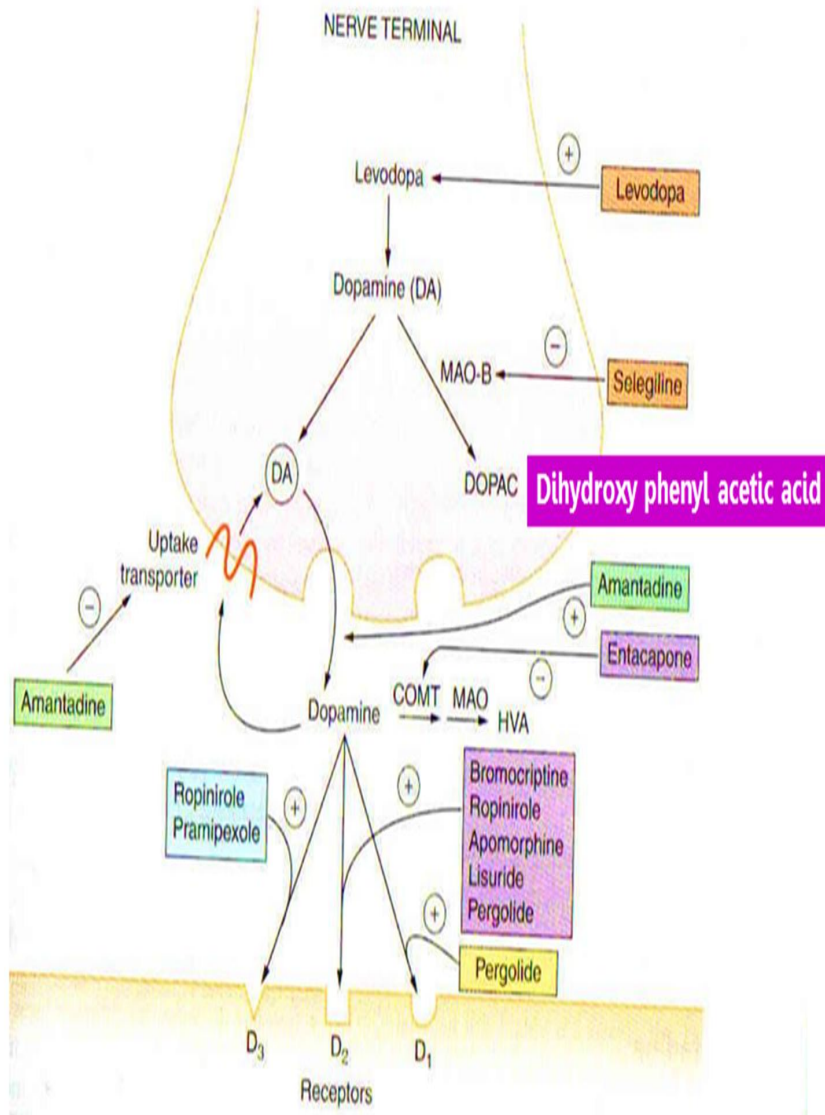
- I. Drugs affecting brain dopaminergic system**

- (a) Dopamine precursor : Levodopa (L-dopa)
- (b) Peripheral decarboxylase inhibitors:
 - ✓ Carbidopa, Benserazide.
- (c) Dopaminergic agonists:
 - ✓ Bromocriptine, Ropinirole, Pramipexole
- (d) MAO-B inhibitor: Selegiline
- (e) COMT inhibitors: Entacapone, Tolcapone
- (f) Dopamine facilitator: Amantadine.

- II. Drugs affecting brain cholinergic system**

- (a) Central anticholinergics: Trihexyphenidyl , Procydidine, Biperiden.
- (b) Antihistaminics : Orphenadrine, Promethazine.

Treatment targets



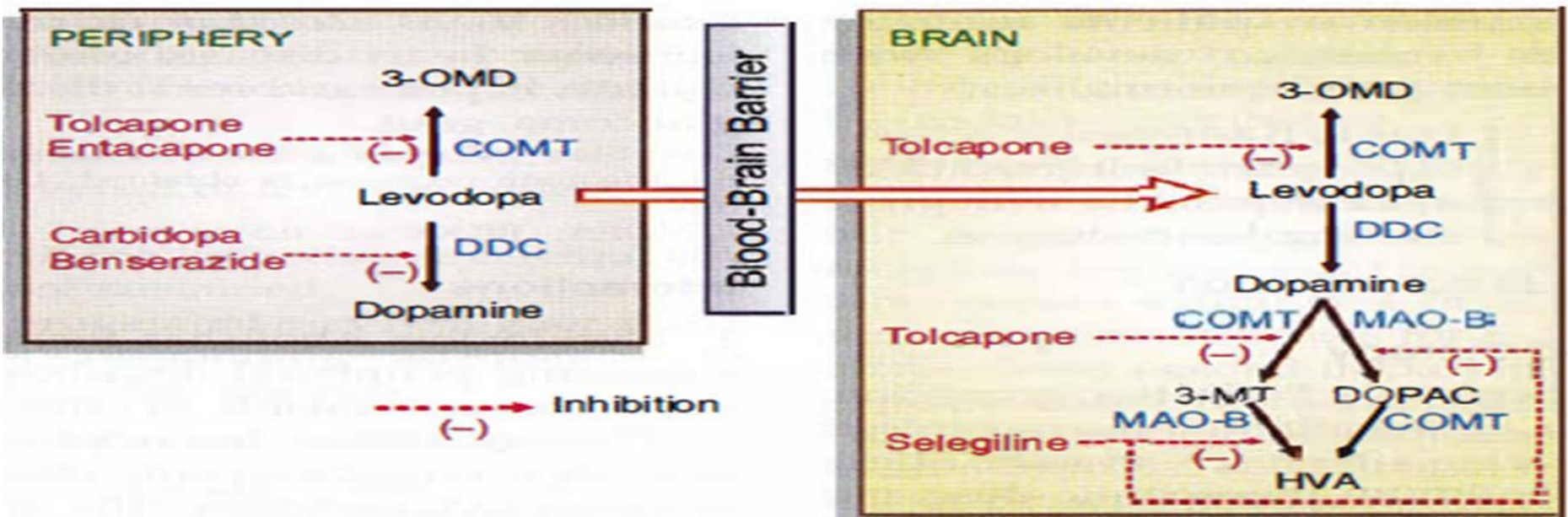
❖ Levodopa and carbidopa

Levodopa

- **MOA**
- it is actively transported into the CNS and is converted to dopamine in the brain
- It decreases the rigidity, tremors, and other symptoms of parkinsonism
- More than 95% of an oral dose is decarboxylated to dopamine in the periphery, So large doses of levodopa are required,
- resulting in side effects that include nausea, vomiting, and cardiac arrhythmias

- **Therapeutic uses**

- Levodopa in combination with carbidopa is a potent and efficacious drug regimen to treat Parkinson's disease.
- levodopa-carbidopa treatment substantially reduces the severity of the disease for the first few years of treatment and then response to the drug is declined



■ Pharmacokinetics

- absorbed rapidly from the small intestine (**when empty of food**)
- has an extremely short half-life (1 to 2 hours)
 - causes fluctuations in plasma concentration which may produce fluctuations in motor response
- Levodopa undergoes high first pass metabolism in g.i. mucosa and liver.

❖ ACTIONS

- 1 . CNS
 - Marked symptomatic improvement
 - ✓ Hypokinesia and rigidity resolve first, later tremor as well, posture, gait, handwriting, speech, facial expression, mood, self care and interest in life are gradually normalized.
- 2.CVS
 - ✓ tachycardia by acting on β_1 adrenergic receptors.
- 3. CTZ
 - ✓ nausea and vomiting.
- 4. Endocrine
 - ✓ DA acts on pituitary to inhibit prolactin release and increase GH release.
 - ✓ prolactin levels in blood fall during levodopa therapy,
 - ✓ but GH levels are not noted in parkinsonian patients.

■ **Adverse effects**

– **Peripheral effects:**

- ✓ Anorexia, nausea, and vomiting occur because of stimulation of the chemoreceptor trigger zone of the medulla.
- ✓ hypotension, tachycardia and hypertension depending on dose
- ✓ mydriasis
- ✓ Saliva and urine are a brownish color because of the melanin pigment produced from catecholamine oxidation.

– **CNS effects**

- Visual and auditory hallucinations
- abnormal involuntary movements (dyskinesias) may occur.
- Levodopa can also cause mood changes, depression, psychosis, and anxiety.

■ Drug interactions

- The vitamin pyridoxine (B6) increases the peripheral breakdown of levodopa and diminishes its effectiveness
- MAOI, such as **phenelzine**, can produce a hypertensive crisis
- **Antipsychotic** drugs are generally contraindicated in parkinsonian patients
- Atropine, and its congeners have additive antiparkinsonian action, but retard its absorption---efficacy of levodopa may be reduced
- Phenothiazines, butyrophenones, metoclopramide reverse therapeutic effect of levodopa by blocking DA receptors

■ **Carbidopa and benserazide**

❖ **MOA**

- Enhances the effects of levodopa on the CNS when coadministered by inhibiting a dopa decarboxylase in the peripheral tissues and GIT
- does not cross the blood-brain barrier
- The addition of carbidopa lowers the dose of levodopa needed by four- to five-fold
- consequently, decreases the severity of the side effects arising from peripherally formed dopamine.

- ***Selegiline***

- *MOA*

- ✓ at low to moderate doses, it *selectively inhibits MAO Type B which* metabolizes dopamine (increase level in the brain)

- Selegiline substantially reduces the required dose of levodopa

- has little potential for causing hypertensive crises

- ✓ at above recommended doses, it inhibit MAO Type A (which metabolizes norepinephrine and serotonin)

- risk for severe hypertension

- Selegiline is metabolized to methamphetamine and amphetamine, whose stimulating properties may produce insomnia if the drug is administered later than **mid afternoon**.
- *Rasagiline*
 - Unlike Selegiline an irreversible and selective inhibitor of brain (MAO) Type B, has five times the potency of selegiline

- ***Catechol-O-methyltransferase (COMT) inhibitors***

- **Entacapone or tolcapone**

- ***MOA***

- selectively and reversibly inhibit COMT and lead to decreased plasma concentrations of 3-O-methyldopa which competes with levodopa for active transport into the CNS
 - Are extensively bound to plasma albumin (>98 %) with limited volumes of distribution

- **Adverse effects**

- including diarrhea, postural hypotension, nausea, anorexia, dyskinesias, hallucinations, and sleep disorders.

- **Tolcapone**

- has a relatively longer duration of action

- Only tolcapone penetrates the blood-brain barrier and inhibits COMT in the CNS

- Most seriously, fulminating hepatic necrosis is associated with tolcapone use.

- So it has been largely replaced by entacapone which lacks such effect

- ***Dopamine-receptor agonists***

- **Ergot derivative** :- Bromocriptine
- **Nonergot drugs**:- ropinirole, pramipexole and rotigotine.

- durations of action longer than that of *levodopa* and, thus, have been effective in patients exhibiting fluctuations in their response to *levodopa*
- are all effective in patients with advanced Parkinson's disease complicated by motor fluctuations and dyskinesias

■ ***Bromocriptine***

- *confusion, delirium, nausea, and orthostatic hypotension are more common, whereas dyskinesia is less prominent.*
- In psychiatric illness, *bromocriptine and levodopa may cause the mental condition to worsen.*
- In patients with peripheral vascular disease, a worsening of the vasospasm occurs

- ***Amantadine***

- is effective in the treatment of influenza
- Also has an antiparkinsonism action

- ***MOA***

- increases the release of dopamine, blockading cholinergic receptors, and inhibiting the N-methyl-D-aspartate (NMDA) type of glutamate receptors
- action at NMDA receptors as the primary action at therapeutic concentrations

- If dopamine release is already at a maximum, *amantadine has no effect*
 - The drug may cause restlessness, agitation, confusion, and hallucinations, and at high doses, it may induce acute toxic psychosis
 - Orthostatic hypotension, urinary retention, peripheral edema, and dry mouth also may occur.
- Amantadine is less efficacious than levodopa, and tolerance develops more readily.

- ***Antimuscarinic agents***
 - *benztropine, trihexyphenidyl, procyclidine, and biperiden*
- are much less efficacious than levodopa and play only an adjuvant role in antiparkinsonism therapy.
- All of these drugs can induce mood changes and produce xerostomia (dryness of the mouth) and visual problems

- **Drugs for Alzheimer disease**

➡ **Alzheimer:-**gradually progressive dementia
affecting both cognition and behavior.

➡ A debilitating neurodegenerative disorder associated with a
rapid cognitive decline

- loss of cortical neurons, particularly cholinergic neurons.

⇒ Clearly differs from the normal aging in that it causes dramatic **loss of synapses, neurons and brain activity** in specific anatomical regions, and results in **massive atrophy and gliosis**.

Neuro pathologic factors

- Decreased **Acetyl cholinesterase** activity
- Selective loss of cholinergic neurons
- Increased accumulation of amyloid plaque
- Increased accumulation of neurofibrillary tangles

➡ The formation of amyloid plaques and neurofibrillary tangles are thought to contribute to the

➤ degradation of the neurons in the brain and the subsequent symptoms of AD

Clinical Manifestation

- Increasing difficulties with **memory** (dementia) and with other **cognitive** functions (executive functions, language, attention, judgement, etc).
- Some patients develop psychotic symptoms
- Overtime, mental functions and activities of daily living are increasingly impaired.
- In the late stages, these individuals become profoundly demented and usually die of intercurrent illnesses
-

Drugs used in AD

- Current therapies aim to either
 - ✓ improve cholinergic transmission within the CNS or
 - ✓ prevent excitotoxic actions resulting from overstimulation of NMDA-glutamate receptors
- **Note:**
 - ✓ Pharmacologic intervention for Alzheimer's disease is only palliative and provides modest short-term benefit.
 - ✓ None of the available therapeutic agents alter the underlying neurodegenerative process.

Acetyl cholinesterase inhibitors:

- The reversible AChE inhibitors approved for the Rx of mild to moderate Alzheimer's disease include **donepezil, galantamine, and rivastigmine**.
 - All of them have some selectivity for AChE in the CNS, as compared to the periphery.
- **Galantamine**
 - augment the action of acetylcholine at nicotinic receptors in the CNS.
 - ✓ modest reduction in the rate of loss of cognitive functioning.

❖ *Rivastigmine*

- is the only agent approved for the management of dementia associated with Parkinson's disease
 - ✓ the only transdermal formulation.
 - ✓ hydrolyzed by AChE to a carbamate metabolite.
- others substrates for CYP450 --potential drug interactions.
- Common adverse effects ; nausea, diarrhea, vomiting, anorexia, tremors, bradycardia, and muscle cramps

- **NMDA receptor antagonist**

- Stimulation of glutamate receptors in the CNS is critical--- certain memories.
- but, overstimulation of glutamate receptors, particularly of the NMDA type, may result in *excitotoxic* effects

✓ *neurodegenerative or apoptotic*

- Binding of glutamate to the NMDA receptor assists in the opening of Ca^{2+} channel that allows to enter the neuron.
- Excess intracellular Ca^{2+} can activate a number of processes that ultimately damage neurons and lead to *apoptosis*.

- **Memantine**

- ✓ NMDA receptor antagonist indicated for moderate to severe Alzheimer's disease.
- ✓ blocking the NMDA receptor and limiting Ca^{2+} influx into the neuron, such that toxic intracellular levels are not achieved.
- ✓ Expected side effects, such as confusion, agitation, and restlessness,
 - indistinguishable from the symptoms of Alzheimer's disease.
- ✓ often given in combination with an AChE inhibitor.

❖ Drugs for Multiple sclerosis disease

- ✓ is an **autoimmune** inflammatory demyelinating disease of the CNS.
- ✓ Historically, corticosteroids used to treat acute exacerbations of the disease.
- ✓ Chemotherapeutic agents, such as *cyclophosphamide* and *azathioprine*, have also been used.
- **Symptomatic treatment**
 - ✓ used to manage of spasticity, constipation, bladder dysfunction, and depression.
 - ***Dalfampridine:***
 - ✓ potassium channel blocker, improves walking speeds in patients with MS.
 - ✓ It is the **first** drug approved for this use.

- **Disease-modifying therapies**

- ✓ decrease relapse rates or prevent accumulation of disability.
- ✓ The major target modify the immune response through inhibition of white blood cell-mediated inflammatory processes

- ❖ **Interferon:**

- ✓ diminish the inflammatory responses that lead to demyelination of the axon sheaths.
- ✓ Adverse effects depression, local injection site reactions, hepatic enzyme increases, and flulike symptoms.
- ✓ Other drugs also available in the same MOA

Antipsychotic Agents

- **Antipsychotics/neuroleptic drugs/major tranquilizers-**
are a group of drugs used mainly for treating schizophrenia
- The term psychosis refers to a **variety** of mental disorders
- characterized by one or more of the following symptoms:
 - ✓ diminished and distorted capacity to process information
 - ✓ hallucinations, delusions,
 - ✓ incoherence or marked loosening of associations,
 - ✓ catatonic or disorganized behavior,
 - ***aggression or violence.***

Psychosis

Psychosis may be part of mood disorder

- **Schizophrenia** (severe thought disorder), Mania (elevated mood), Depression, Toxic psychosis (drug induced)
- **Schizophrenia:** is a chronic psychotic illness
 - The main clinical features
 - **Positive symptoms**
 - ✓ Delusions, hallucinations, usually in the form of voices, thought disorder eg garbled sentences, irrational conclusions, abnormal behaviours, such as aggressive behaviours.
 - **Negative symptoms:**
 - ✓ withdrawal from social contacts, flattening of emotional responses.

■ Etiology

- The cause remains unclear but involves a combination of genetic and environmental factors

① Genetic & Environmental factors

- has a strong genetic component and probably reflects a **dysfunction** of the mesolimbic or mesocortical dopaminergic neurons.
- Some environmental factors have been identified as possible predisposing factors (E.g. maternal viral infections)
- **Possible primary defects include**
 - excessive activation of CNS receptors for **dopamine**

Etiology...

② Neuro-chemical theories

➡ Dopamine hypothesis of schizophrenia suggests that schizophrenia results from excess activity at dopamine synapses in certain areas of the brain.

➡ Glutamate hypothesis of schizophrenia suggests the problem relates partially to deficient activity at glutamate receptors.

- Especially in the prefrontal cortex

➡ Serotonin hypothesis of schizophrenia suggests that results from excess activity of 5-HT.

- **The goals of treatment**

- are to reduce symptoms, decrease psychotic relapses, and improve patient functioning and social outcomes.

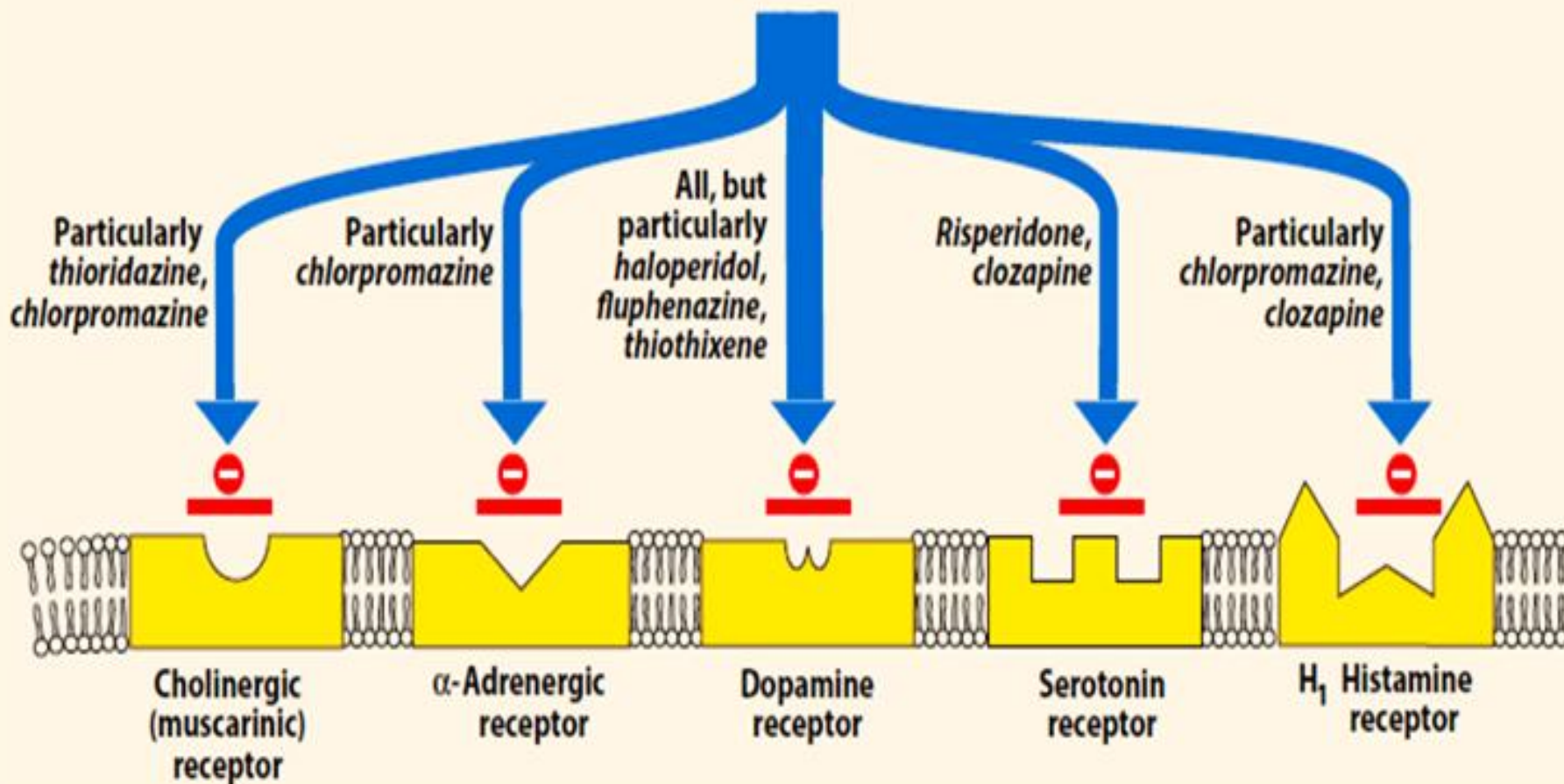
- most patients with schizophrenia relapse when not medicated, so long-term treatment is usually necessary.

- Mechanism of action

- All antipsychotics (except clozapine-like atypical) have potent dopamine D₂ receptor blocking action

Actions of anti psychotics

ANTIPSYCHOTIC DRUGS



❖ Anti psychotic drugs

- **Traditional/typical /first generation /conventional**
 - chlorpromazine, thioridazine, haloperidol, flupenthixol
 - chlorpromazine was the first Anti psychotic drug used to treat schizophrenia
 - are competitive inhibitors at a variety of receptors but their antipsychotic effects reflect competitive blocking of dopamine receptors(D2) with longer duration
 - Hence, Extra Pyramidal Side Effects (less dopamine, higher (unbalanced) Ach)⇒**Affect motor & movement**
 - vary in potency (low, medium, or high potency.), but No one drug is clinically more effective than another

- **Thioridazine**

- ✓ having marked central anticholinergic action.
- ✓ Incidence of extrapyramidal side effects is very low.
- ✓ Cardiac arrhythmias and interference with male sexual function are more common.
- ✓ Risk of eye damage limits long-term use
- ✓ **Thioridazine & chlorpromazine :↑ H1,M& α block & ↑ weight gain**

- **Haloperidol**

- ✓ produces few autonomic effects, is less epileptogenic, does not cause weight gain, jaundice is rare.
- ✓ It is the preferred drug for acute schizophrenia.

■ **Mechanism of action**

➤ **Dopamine blockade**

- All of the older and most of the newer neuroleptic drugs block dopamine receptors in the brain and the periphery
- ✓ There are 5 types of Dopamine receptors (D1-D5)
- ✓ But the clinical efficacy of the typical neuroleptic drugs correlates closely with their relative ability to block D2 receptors in the mesolimbic system of the brain.

However, many of antipsychotic agents also block cholinergic, adrenergic, and histaminergic receptors

- The undesirable side effects of these agents, however, are often a result of actions at these other receptors.

- All of the neuroleptic drugs can reduce the hallucinations and delusions associated with schizophrenia (the so-called positive symptoms)
 - negative symptoms, such as anhedonia are not as responsive to therapy, particularly with the typical neuroleptics.
 - » Many atypical agents, such as clozapine, ameliorate the negative symptoms to some extent

- The antipsychotic effects usually take several days to weeks to occur
 - ✓ Most antipsychotic drugs are readily but incompletely absorbed when given orally.
 - ✓ Some are given as IM injection
 - ✓ Many of these drugs undergo significant first-pass metabolism.
 - ✓ their therapeutic index is high

❖ Action of Traditional Anti psychotic drugs

- ✓ **CNS:** reduces irrational behaviour, agitation and aggressiveness and controls psychotic symptomatology
- ✓ **ANS:** blocking activity Of α adrenergic
- ✓ **Local anaesthetic:** Chlorpromazine is as potent a local anaesthetic as procaine. But not used ---*irritant action*.
 - ✓ Others have weaker membrane stabilizing action.
- ✓ **CVS:** hypotension by central & peripheral action on sympathetic
- ✓ **Endocrine:** consistently increase prolactin release by blocking the inhibitory action of DA on pituitary
 - ✓ This may result in galactorrhoea and gynaecomastia

❖ Atypical (2nd generation, modern drugs)

➡ Strong 5H2A antagonists; weak & transient D2 antagonists

➡ Characteristics features of atypical antipsychotics

- Less (none) EPS and tardive dyskinesia
 - Lack of effect on serum prolactin
 - Greater efficacy for **refractory** schizophrenia
 - Greater activity against **negative** symptoms
- **Clozapine** is actually the only atypical agent which fulfills all of the criteria stated above.
- Atypicals associated with a higher risk of metabolic side effects, such as Diabetes, hypercholesterolemia, and weight gain

❖ Olanzapine

- This atypical antipsychotic; resembles clozapine in blocking multiple monoaminergic (D2, 5-HT2I, α 1,2) as well as muscarinic and HI receptors.
- Both positive and negative symptoms of schizophrenia appear to be benefited.
- A broader spectrum of efficacy covering schizo-affective disorders, and it is approved for use in mania.
 - ✓ Monotherapy with olanzapine may be as effective as a combination of lithium/valproate + benzodiazepines.

Therapeutic Uses of antipsychotics

➡ Treatment of schizophrenia

- to suppress symptoms and help prevent relapse

➡ Prevention of severe nausea and vomiting

- older neuroleptics (most commonly prochlorperazine) are useful in the treatment of drug-induced nausea

➡ for treatment of chronic pain with severe anxiety

- Neuroleptics are used in combination with narcotic analgesics

➡ Chlorpromazine is used to treat intractable hiccups

- ❖ extrapyramidal **side effects** occur with chronic treatment
 - ✓ **dystonias** (sustained contraction of muscles leading to twisting distorted postures)
 - ✓ **akathisia** (motor restlessness), and
 - ✓ **tardive dyskinesia** (involuntary movements of the tongue, lips, neck, trunk, and limbs)
- Blocking of dopamine receptors in the nigrostriatal pathway probably causes these unwanted movement symptoms.
- The atypical neuroleptics exhibit a lower incidence of these symptoms.

Summery antipsychotics relative potency & side effect

Drug	oral dose (mg)	Extrapyramidal effects	Sedation	Orthostatic hypotension	Anticholinergic effects
Conventional					
<i>Low potency</i>					
Chlorpromazine	100	Moderate	High	High	Moderate
Thioridazine	100	Low	High	High	High
<i>Moderate potency</i>					
	25	Moderate	High	Moderate	Moderate
Triflupromazine	10	Moderate	Moderate	Low	Low
Perphenazine	10	Moderate	Moderate	Low	Low
Loxapine					
<i>High potency</i>					
	2	High	Low	Low	Low
Haloperidol	2	High	Low	Low	Low
Fluphenazine					
Atypical Agents					
Clozapine	50	Very low	High	Moderate	High
Risperidone	4	Very low	Low	Low	None

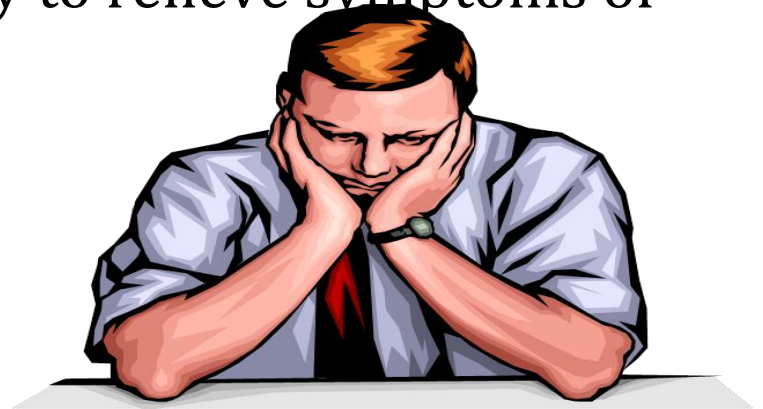
- Neuroleptics can also aggravate preexisting epilepsy, and they should be used with caution in patients with epilepsy
 - chlorpromazine and clozapine are contraindicated in patients with seizure disorders
- high incidence of agranulocytosis with clozapine may limit its use to patients who are resistant to other drugs.
- Weight gain commonly occurs with atypical neuroleptics

- **INTERACTIONS**

- 1. Neuroleptics potentiate all CNS depressants-hypnotics, anxiolytics, alcohol, opioids, antihistaminics and analgesics--
-----*Overdose symptoms may occur.*
- 2. Neuroleptics block the actions of levodopa and direct DA agonists in parkinsonism.
- 3. Antihypertensive action of clonidine and methyldopa is reduced, probably due to central α_2 adrenergic blockade.
- 4. Enzyme inducers (barbiturates, anticonvulsants) can reduce blood levels of neuroleptics.

Antidepressants

- Antidepressants are used primarily to relieve symptoms of depression.
- **Clinical Features of depression:**
 - The symptoms of depression are intense feelings of **sadness, hopelessness**, as well as the **inability** to experience pleasure in usual activities, **changes in sleep patterns** and **appetite** (poor), **loss of energy**, and **suicidal thoughts**



Etiology & Pathogenesis

① Genetic causes

② Environmental factors

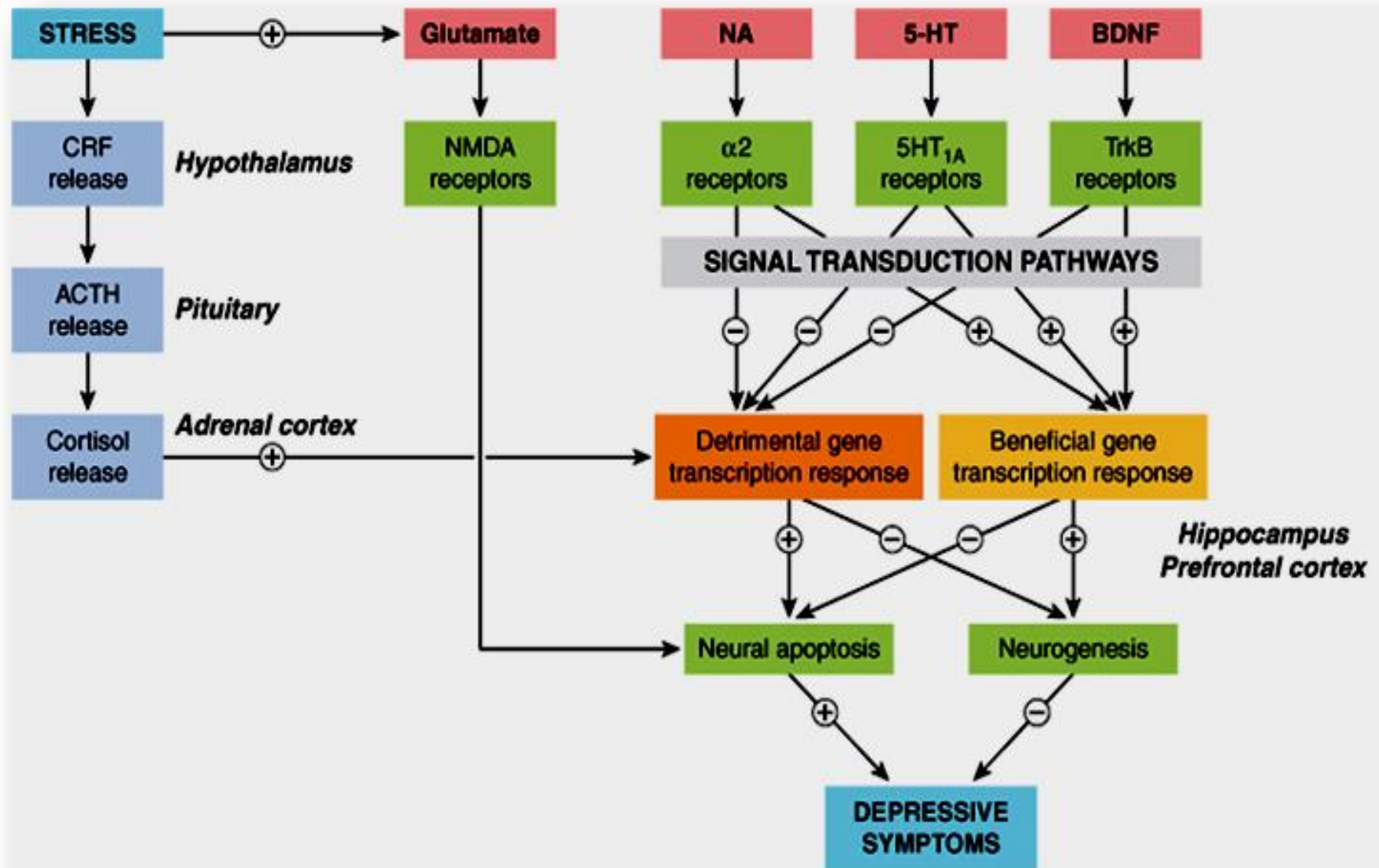
③ Biochemical factors

- Deficiency of NT amines in certain part of the brain (NA, 5-HT & DA)

④ Neuro-Endocrine factors

- ↓BDNF & ↑ plasma cortisol level in depressed patients
- Abnormality in brain derived neurotrophic factor (BDNF) is associated with depression
 - ✓ BDNF are critical in the regulation of neural **plasticity**, **resilience**, and **neurogenesis**

Depressive pathways



■ **Treatment Modalities**

- ✓ Pharmacotherapy
- ✓ depression-specific psychotherapy
- ✓ electroconvulsive therapy

■ **Antidepressant drugs**

– The types of antidepressants are:

- ✓ Tricyclic antidepressants (TCAs)
- ✓ Monoamine oxidase inhibitors (MAOIs)
- ✓ Selective serotonin reuptake inhibitors (SSRIs)
- ✓ serotonin/norepinephrine reuptake inhibitors (SNRIS)
- ✓ atypical antidepressants eg bupropion

■ **mechanism of antidepressant**

- ✓ Most clinically useful antidepressant drugs potentiate, either **directly** or **indirectly**, the actions of norepinephrine and/or serotonin in the brain
- ✓ Most require at least 2 weeks for beginning effect/mood improvement
- ✓ NB: all antidepressants may lower the seizure threshold
- ✓ NB: These drugs do not usually produce central nervous system (CNS) stimulation or mood elevation in normal individuals

1. Tricyclic Antidepressants (TCAs)

- ✓ Amitryptiline, nortryptiline ,Imipramine, desipramine, Doxepin and amoxapine Were **first-choice** drugs
- *Mechanism of Action*
 - block the uptake of amines (noradrenaline and 5-HT) by nerve terminals by competition for the carrier transport system
 - also block $\alpha 1$ - adrenoceptors, muscarinic, histamine (H1) and 5-HT receptors.

■ Pharmacokinetics

- Long and variable half-lives; given in single daily dose
- Most tricyclics are incompletely absorbed and undergo significant first-pass metabolism
- Highly protein bound and relatively high lipid solubility.
- The TCAs have a narrow therapeutic index

■ **Therapeutic Uses**

👉 Treating moderate to severe depression

1. Depression—Takes 1–3 weeks for initial responses

— are used to treat episodes of major depression

2. Bipolar disorder, chronic insomnia, attention-deficit/hyperactivity disorders, and panic disorder

👉 Bed-wetting in children (> 6 yrs) (imipramine)

- (contraction of the internal sphincter of the bladder)

👉 To treat migraine headache and chronic pain syndromes (neuropathic pain) (amitriptyline)

👉 Insomnia (doxepin)

Adverse effects:

- ☠️ **Antimuscarinic:** blurred vision, xerostomia (dry mouth), urinary retention, sinus tachycardia, constipation, and aggravation of narrow-angle glaucoma
- ☠️ **Block α -adrenergic receptors:** orthostatic hypotension, dizziness, and reflex tachycardia
- ☠️ Erectile dysfunction in men and anorgasmia in women
- ☠️ **Block histamine H₁ receptors:** sedation, weight gain
- ☠️ **Note:** The TCAs have a significant risk for arrhythmias in overdose
- ☠️ Abrupt withdrawal of TCAs (especially high doses) may result in cholinergic rebound (eg, dizziness, nausea, diarrhea, insomnia, and restlessness)

■ Comparative properties of tricyclic antidepressants

Drug	Sedation	Anti-muscarinic	Hypotension	Cardiac arrhythmia	Seizure precipitation	Daily dose (mg)
<i>Tricyclic antidepressants</i>						
1. Imipramine	+	++	++	+++	++	50–200
2. Amitriptyline	+++	+++	+++	+++	++	50–200
3. Trimipramine	+++	+++	++	+++	++	50–150
4. Doxepin	+++	++	++	++	++	50–150
5. Clomipramine	++	+++	++	+++	+++	50–150
6. Dothiepin (Dosulpin)	++	++	++	++	++	50–150
7. Nortriptyline	+	++	+	++	+	50–150
8. Amoxapine	+	+	++	++	++	100–300

■ *Drug interactions*

- increase the catecholamine effects of amphetamines and sympathomimetics, leading to **hypertension**.
 - Barbiturates increase the metabolism of TCAs
 - TCAs with MAOIs may cause an extremely elevated body temperature, excitation, and seizures.

2. Selective Serotonin Reuptake Inhibitors (SSRIs)

- fluoxetine, fluvoxamine, paroxetine, sertraline, Citalopram

☛ **MOA:-**Specifically inhibit serotonin reuptake

☛ have 300 to 3000-fold greater selectivity for the serotonin transporter, as compared to the Norepinephrine transporter

☛ Have little blocking activity at muscarinic, α -adrenergic, and histaminic H₁ receptors

⇒ Common side effects associated with TCAs, such as orthostatic hypotension, sedation, dry mouth, and blurred vision, are not commonly seen with the SSRIs

➤ The drugs of choice in treating depression

■ Clinical use

- the SSRIs have largely replaced tricyclic antidepressants and monoamine oxidase inhibitors as the drugs of choice in treating depression.
 - » Because they have fewer adverse effects and are relatively safe even in overdose
- generalized anxiety disorder, panic disorder, posttraumatic stress disorder, phobias, social anxiety disorder, premenstrual dysphoric disorder, bulimia nervosa(only fluoxetine), obsessive-compulsive disorder (fluvoxamine)

Pharmacokinetics

- All of the SSRIs are well absorbed after oral administration
- Food has little effect on absorption (except with **sertraline**, for which food increases its absorption)
- The majority of SSRIs have plasma half-lives that range between 16 and 36 hours
- Metabolism by CYP450-dependent enzymes and glucuronide or sulfate conjugation occur extensively

Fluoxetine

- Differs from the others by having a much longer half-life
- Half life of its active metabolite S-norfluoxetine is ≈ 10 days
- It is available as a sustained-release preparation allowing once-weekly dosing.
- Fluoxetine and paroxetine are potent inhibitors of CYP2D6 (some extent others) responsible for the elimination of TCAs, antipsychotic drugs....
- Dosages of the SSRIs should be reduced in patients with hepatic impairment

- **Drug interactions**

- competitively inhibit a liver enzyme
- don't mix with MAOIs

- **Adverse reactions to SSRIs**

- Anxiety, insomnia, somnolence, palpitations, various rashes, changes in weight and sexual dysfunction (eg delayed ejaculation)
- Withdrawal syndrome following discontinuance in depression: nausea, headache, vertigo, malaise.

- **Serotonin-Norepinephrine Reuptake Inhibitors**

Venlafaxine and duloxetine

- **MOA** - selectively inhibit the re-uptake of both serotonin and norepinephrine
 - may be effective in treating depression in patients in whom SSRIs are ineffective
 - may precipitate a discontinuation syndrome if treatment is abruptly stopped.

3. Monoamine Oxidase Inhibitors (MAOIs)

- Selegiline, Phenelzine, Tranylcypromine
 - Use of MAO inhibitors is now limited due to the complicated dietary restrictions required of patients taking MAO inhibitors.
- **MOA**
 - Most MAO inhibitors, such as phenelzine cause irreversible inactivation of the enzyme
 - Therefore ; increased stores of NE, 5-HT, and dopamine within the neuron

- are well absorbed after oral administration
- are metabolized in the liver, excreted mainly by the GI tract
- ✓ Used as second- and third-choice drugs especially since major problem is hypertensive crisis with tyramine foods.
- ✓ indicated for depressed patients who are unresponsive or allergic to TCAs or who experience strong anxiety.
- ✓ also useful in the treatment of phobic states
- ✓ atypical depression, may respond to MAO inhibitors

- when switching to another antidepressant agents, a minimum of 2 weeks of delay must be allowed after termination of MAO inhibitor therapy

■ **Adverse effects** : include drowsiness, orthostatic hypotension, blurred vision, dry mouth, dysuria, and constipation

- MAOI cause excessive central stimulation and weight gain.
- Adverse reactions may be avoided by giving the drug in small divided doses

- Certain foods can interact with MAOIs and produce severe reactions.
 - The most serious reactions involve tyramine -rich foods, such as red wines, aged cheese
 - Foods with moderate tyramine contents for example, ripe banana may be eaten occasionally, but with care

Treatment of Mania and Bipolar Disorder

- Mania
 - the patient experiences excessive catecholamine stimulation.
- bipolar disorder
 - the patient is affected by swings between the excessive catecholamine stimulation of mania and the diminished catecholamine stimulation of depression.

- **Lithium**

- Its salts are used **prophylactically** for treating manic-depressive patients and in the **treatment** of manic episodes

- **MOA**

- exact mechanism of action is unknown
 - may regulate catecholamine release in the CNS by:
 - increasing norepinephrine and serotonin uptake
 - reducing the release of NE
 - inhibiting norepinephrine's action in the postsynaptic neuron.

- When taken orally, lithium is absorbed rapidly and completely and is distributed to body tissues.
- Slow onset (2 weeks)
- Lithium has a **narrow** therapeutic margin of safety.

■ **Clinical use**

- Lithium is used primarily to treat acute episodes of mania and to prevent relapses of bipolar disorders.

■ Adverse effects

- Common adverse reactions to lithium include:
 - ✓ headache, dry mouth, polydipsia, polyuria, polyphagia, gastrointestinal distress (give lithium with food), fine hand tremor, dizziness, fatigue, dermatologic reactions, and sedation.
 - Thyroid function may be decreased

■ Drug interactions

- Eg: risk of lithium toxicity increases when lithium is taken with **thiazide** and **loop** diuretics and **non-steroidal** anti-inflammatory drugs

- Several antiepileptic drugs, including most notably carbamazepine, valproic acid, and lamotrigine have been successfully utilized in the treatment of bipolar disorder
- Other agents that may improve manic symptoms include the **older** and **newer** antipsychotics
- Benzodiazepines are also frequently used as adjunctive treatments for the acute stabilization of patients with mania

Pharmacology of drugs used to treat pain

▪ Introduction

- **Pain** : is an unpleasant sensory and emotional experience that is associated with actual or potential tissue damage
- Pain stimuli are detected by physiological receptors(nociceptors)
- Small pain conducting fibers transmit the signal of injury to **thalamus** and **limbic** system
- This transmission is regulated (reduced) by large mechanofibers

- **Neurotransmitters involved in pain pathways**
 - include acetylcholine, histamine, polypeptides such as bradykinin, prostaglandin(PG) , and substance P.
- **Acute pain**
 - is of short duration and lasts less than 3 to 6 months.
 - Eg postoperative pain and traumatic pain.

- **Chronic pain**

- lasts longer than 6 months

- Chronic pain associated with malignant disease includes the pain of cancer, AIDS etc

- Chronic pain not associated with malignant disease includes headaches, rheumatoid arthritis etc

- **Analgesic drug:**
 - Is a drug that relieves pain due to multiple causes, e.g. paracetamol, morphine.
- Can be broadly classified as
 - **Non narcotic analgesics**
 - salicylates
 - nonsalicylates [acetaminophen]
 - the nonsteroidal anti-inflammatory drugs
 - **the narcotic analgesics**

- **Non narcotic analgesics**

- Unlike narcotic analgesics, these drugs are used to relieve pain **without** the possibility of *causing physical dependency*
- Can be divided into the salicylates, nonsalicylates (acetaminophen), and the nonsteroidal anti-inflammatory drugs (NSAIDs)
- combinations of nonnarcotic analgesics are available

- **Salicylates**

- aspirin (acetylsalicylic acid), magnesium salicylate, sodium salicylate etc
 - have analgesic (relieves pain), antipyretic (reduces elevated body temperature), and anti-inflammatory effects
- aspirin
 - » It is the most commonly used salicylate
 - » has a greater anti-inflammatory effect than the other salicylates
 - » also prolongs the bleeding time by inhibiting the aggregation (clumping) of platelets. other salicylates do not have as great an effect on platelets as does aspirin

■ MOA

- due to the inhibition of prostaglandins
 - » PGs increase the sensitivity of peripheral pain receptors
- Aspirin is irreversible inhibitor of cyclooxygenase enzyme
- Aspirin can be given orally and rectally
- Maximum dose is 8g/d

■ **Therapeutic uses**

- Salicylate analgesics are used for the following reasons:
 - Relief of mild to moderate pain
 - Reduction of elevated body temperature
 - Treatment of inflammatory conditions, such as rheumatoid arthritis, osteoarthritis, and rheumatic fever
 - Aspirin only – Reduction of the risk of myocardial infarction and transient ischemic attacks
- Aspirin and other NSAIDs have been combined with opioid analgesics for treatment of cancer pain.

■ Adverse effects

- Gastric upset, heartburn, nausea, vomiting, anorexia, and gastrointestinal bleeding
- use of these drugs over a long period, even in normal doses, can result in a significant blood loss
- Allergic reactions
 - urticaria, rash, angioedema, bronchospasm, and anaphylactoid reactions.
- Salicylate toxicity produces a condition called salicylism (tinnitus, dizziness, fever, headache, hyperventilation)
 - » Mild salicylism is reversible with reduction of the drug dosage.

■ **Contraindications**

- known hypersensitivity to the salicylates or the NSAIDs
- during pregnancy particularly third trimester
- breast-feeding
- in those with bleeding disorders or bleeding tendencies
- Children or teenagers with influenza or chickenpox should not take the salicylates, particularly aspirin
 - » If taken by such patients, they cause Reye's syndrome (a life threatening condition characterized by vomiting and lethargy, progressing to coma)

- **Nonsalicylates eg . acetaminophen**

- The major drug classified as a nonsalicylate is acetaminophen
- has analgesic and antipyretic activity but no antiinflammatory action

- **MOA**

- Reducing pain- may work in the central nervous system (CNS) by inhibiting prostaglandin synthesis and in the peripheral nervous system in some unknown way.
- It reduces fever by acting directly on the heat-regulating center in the hypothalamus.

- It is the most widely used as aspirin substitute as analgesic (mild to moderate) and antipyretic
 - » for patients who are allergic to aspirin
 - » who experience extreme gastric upset when taking aspirin
 - » children with fever and flu-like symptoms

■ Adverse effects

- occur with chronic use or when the recommended dosage is exceeded
- Include skin eruptions, urticaria (hives), hemolytic anemia, hypoglycemia, jaundice (yellow discoloration of the skin), hepatotoxicity and hepatic failure (seen in chronic alcohol)
- With excessive dosages the liver cells **necrose or die.**
- Death can occur due to liver failure
- total daily dose shouldn't exceed 4,000 mg/day

■ **Contraindications**

- Hypersensitivity to acetaminophen
- used cautiously during pregnancy and lactation.
- If an analgesic is necessary, it appears safe for short-term use

Nonsteroidal Anti-inflammatory Drugs (NSAIDs)

- the chemical and physiologic effects are similar to salicylates
 - have analgesic, antipyretic , and anti-inflammatory effects
 - thought to act by inhibiting prostaglandin synthesis by inhibiting the action of the enzyme cyclooxygenase
- 1. cyclooxygenase-1 (COX-1)** - the enzyme that helps to maintain the stomach lining; and
 - 2. cyclooxygenase-2 (COX-2)** - the enzyme that triggers pain and inflammation.

- **The nonselective NSAIDs**

- Eg diclofenac, ibuprofen, indomethacin, naproxen, piroxicam, and sulindac.
- selective NSAIDs- eg celecoxib

- **Therapeutic uses**

- NSAIDs have a variety of uses
 - Generally, they can be used for
 - Relief of signs and symptoms of osteoarthritis, rheumatoid arthritis, and other musculoskeletal disorders
 - ✓ Mild to moderate pain relief
 - ✓ Fever reduction

■ Adverse effects

- Gastrointestinal tract—nausea, vomiting, diarrhea, constipation, epigastric pain, intestinal ulceration, stomatitis etc
- Central nervous system—dizziness, anxiety, lightheadedness, vertigo, headache etc
- Cardiovascular—congestive heart failure, BP change
- Renal—hematuria
- visual disturbances
- Hematologic— aplastic anemia etc
- Skin—rash, erythema etc

■ **Contraindications**

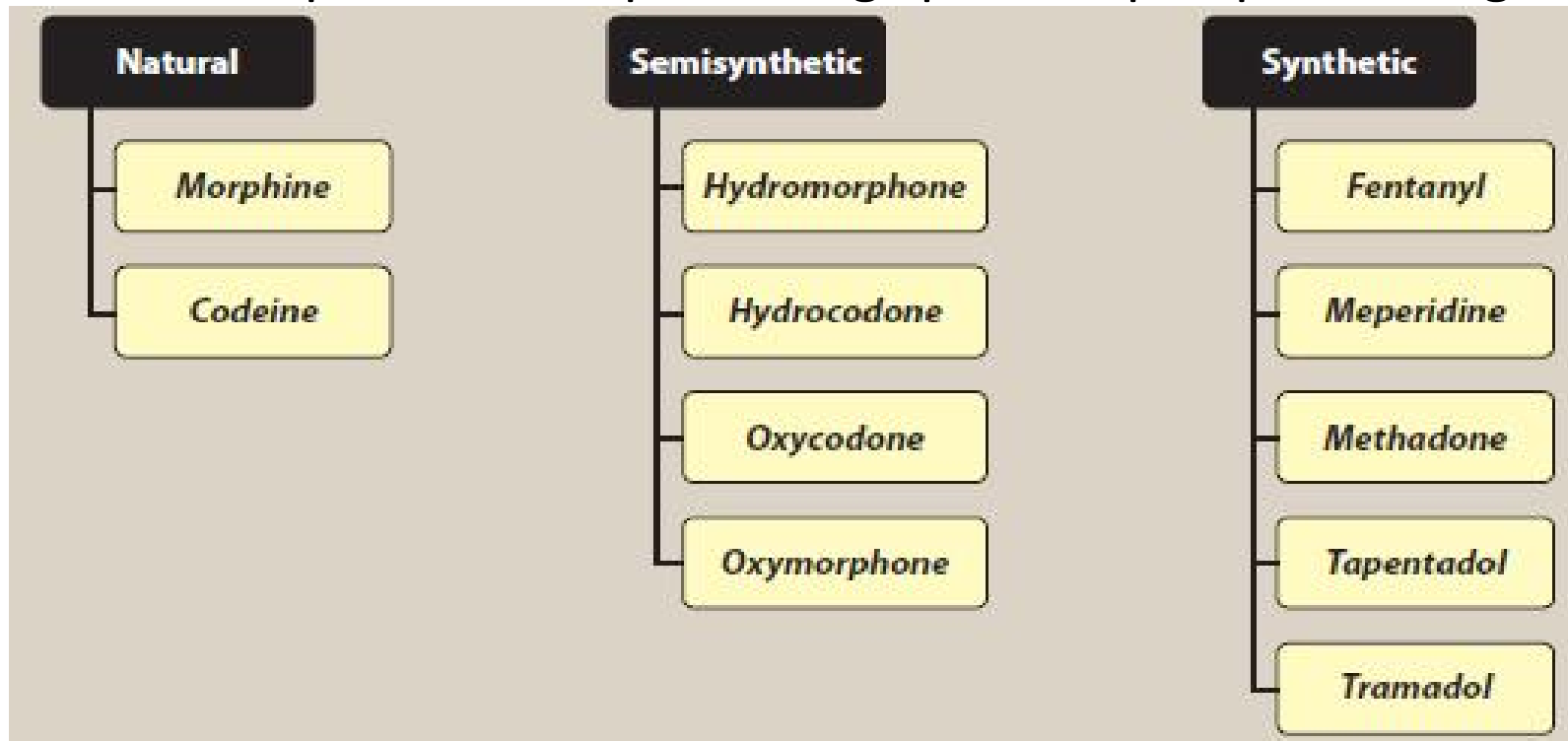
- patients with known hypersensitivity
 - » Hypersensitivity to aspirin is a contraindication for all NSAIDs.
- NSAIDs are contraindicated during the third trimester of pregnancy and during lactation.

❖ Narcotic Analgesics

- Opioid is any drug, natural or synthetic, with actions like morphine
- are controlled substances used to treat moderate to severe pain.
- **enkephalins, endorphins, and dynorphins** serve as neurotransmitters.
- Although five categories of opioid receptors have been identified, **only three** of these receptors affect the action of the narcotic analgesics:
- Narcotic analgesics are classified as agonists, partial agonists, and mixed agonists-antagonists
- **Analgesic** A drug that selectively relieves pain by acting in the **CNS** or on **peripheral** pain mechanisms, without significantly altering consciousness.

Opioid receptors

- **Mu** -activation includes analgesia, respiratory depression, **euphoria**, miosis and constipation.
- **Kappa** -activation produces analgesia, sedation, and **dysphoria** (a state of dissatisfaction).
- **Delta** -are responsible for producing spinal/supraspinal analgesia



■ **Classification of Drugs That Act at Opioid Receptors**

1. Pure opioid agonists

all have high affinity to **mu receptors** and varying affinity to **delta** and **kappa** receptors

Eg Morphine, codeine, methadone, dextropropoxyphene

2. Agonist-antagonists opioids—eg. pentazocine, nalbuphine, butorphanol, and buprenorphine

- ✓ when administered alone, produce analgesia,
 - ✓ but given to a person taking opioids, produce an antagonistic effect.

3. Pure opioid antagonists—Naloxone, naltrexone.

Antagonize mu and kappa receptors and are used for reversal of respiratory and CNS depression.

■ Therapeutic uses

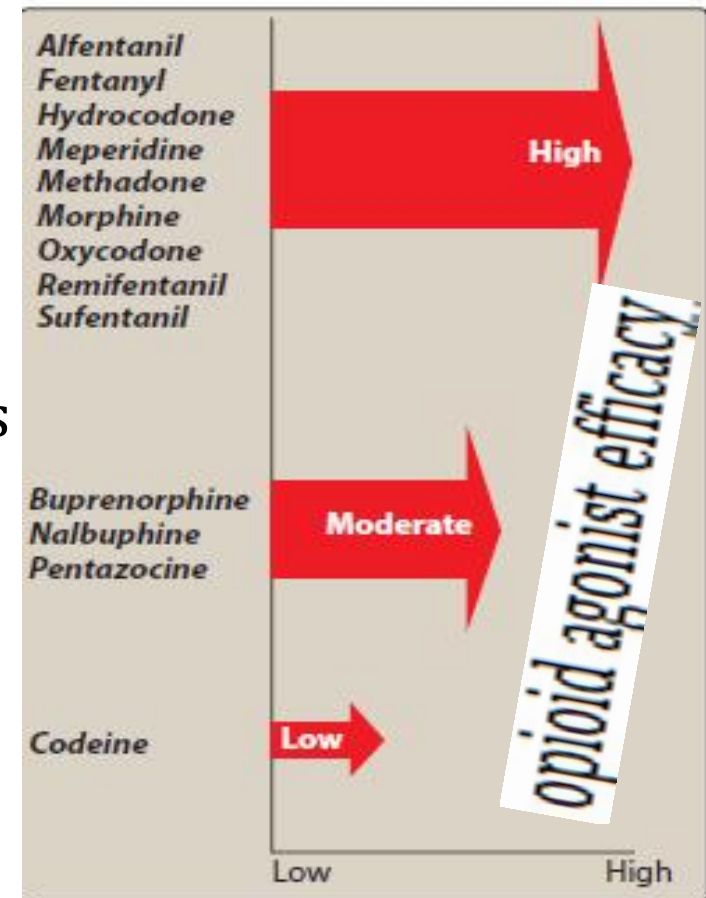
- major use of the narcotic analgesic is to relieve or manage **moderate to severe** acute and chronic pain
 - » Morphine is considered the prototype
 - » It is the most widely used opioid
 - » **codeine** or **pentazocine** can be used
 - For **mild to moderate** pain-
 - severe diarrhea and intestinal cramping
 - Relief of severe, persistent cough
- as an adjunct during anesthesia
 - ✓ To lessen anxiety and sedate the patient before surgery

➡ Morphine:

➤ Produces analgesia, sedation, euphoria, respiratory depression, cough suppression, and decreased bowel motility.

▪ *Mechanism of analgesic action*

- ✓ Mimics endogenous opioid peptides
 - ✓ (mu and kappa receptors).
- undergoes first-pass metabolism
- Can be given PO, IM, IV, SC and rectally



opioid agonist efficacy

- ***Therapeutic use***

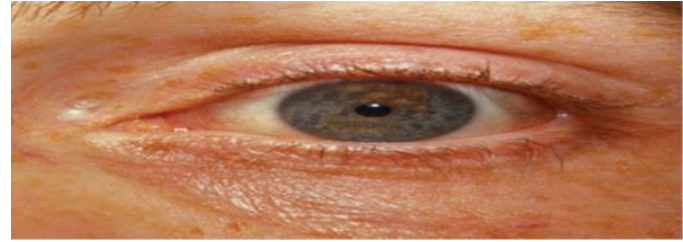
- Relief of pain (moderate to severe) for labor and delivery, MI pain and dyspnea associated with LVF and pulmonary edema

- preoperatively for sedation and anti-anxiety

- ***Adverse effects***

- Respiratory depression, constipation, orthostatic hypotension, urinary retention, cough suppression, **emesis**, increased intracranial pressure (ICP), sedation, euphoria/dysphoria, and miosis
 - Abuse liability

- ***Toxicity***



- » Triad of signs: coma, respiratory depression and pinpoint pupils—***provide support and give IV naloxone.***

- **Tolerance**

- marked for the CNS actions including analgesia, euphoria, and respiratory depression, but it occurs minimally in terms of miosis and the effects on GI motility.
 - Cross-tolerance occurs between individual opioid analgesics.

- **Codeine**

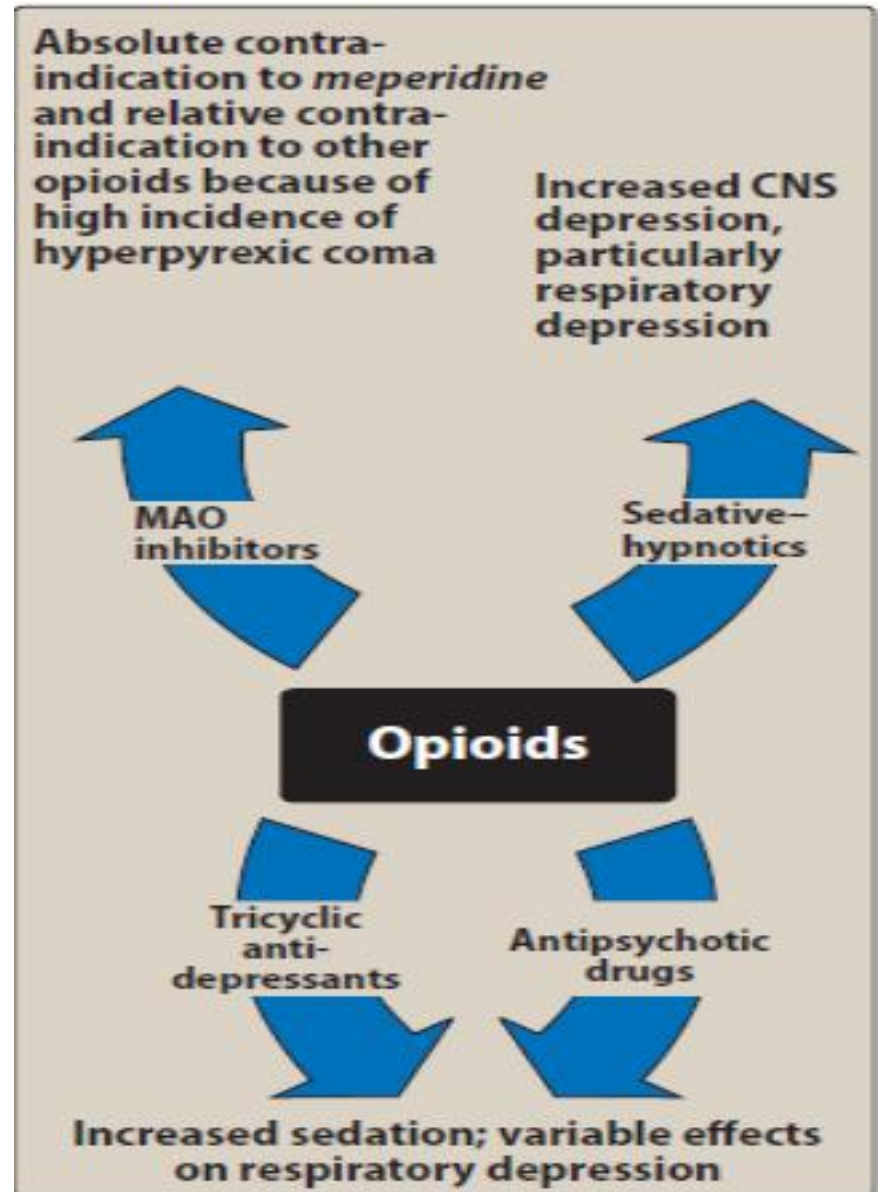
- ✓ weak analgesic compared to *morphine*.
- ✓ used only for mild to moderate pain.
- ✓ The analgesic actions of *codeine* by its conversion to *morphine* by the CYP450 2D6
- ✓ Drug interactions associated with the CYP450 2D6 may alter the ***efficacy of codeine*** or ***potentially lead to toxicity***.
- ✓ commonly used in combination with *acetaminophen*
- ✓ *codeine* has been replaced by *dextromethorphan*, ----***cough***
 - relatively no analgesic action and low potential for abuse

- **Fentanyl**

- ✓ synthetic and highly potent than of *morphine*
- ✓ highly lipophilic and has a rapid onset and short duration of action.
- ✓ combined with local anesthetics to provide epidural analgesia for labor and postoperative pain.
- ✓ used in **anesthesia** for its analgesic and sedative effects.
- ✓ used in the treatment of cancer patients with break through pain who are tolerant to opioids.
- ✓ The transdermal patch must be used with caution because death-----***resulting from hypoventilation***

Drug interactions:

- phenothiazines,
- monoamine oxidase inhibitors (MAOIs),
- tricyclic antidepressants



- **Opioid antagonists**

- Naloxone and naltrexone

- Competitively block the effects of opioid drugs, enkephalins, and endorphins.

- Naloxone is administered IM, SC or IV

- » the drug of choice for managing an opioid overdose

- Naltrexone is administered orally

- Naltrexone is used along with psychotherapy or counseling to treat drug abuse

- **Opioid dependence**

- Tolerance and physical dependence occur rapidly for opioids
- Even patients taking opioids for 3 days can experience withdrawal symptoms

- » Eg mild tremor, sweating, fever,

- Withdrawal from chronic use

- » Increased respiration, perspiration, mydriasis etc

- Two opioids are used in the treatment and management of opiate dependence: levomethadyl and methadone.

■ Anesthesia

✓ **anesthesia** is a loss of feeling or sensation

- The chemical agents can be divided into three groups general anesthetics, local anesthetics, and topical anesthetics.

■ General anesthetics

- produce unconsciousness and lack of responsiveness to painful stimuli
- **MOA**- most general anesthetics enhance the effects of GABA
- **N2O and ketamine** selectively inhibit the excitatory NMDA type of glutamate receptor
- No single agent is perfect. Hence a combination is used.
- **thiopental** for rapid induction, neuromuscular blocker for muscle relaxation

General anesthetics...

- For patients undergoing surgical or medical procedures, anesthesia provides five important benefits:
 - ☞ Sedation and reduced anxiety
 - ☞ Lack of awareness and amnesia
 - ☞ Skeletal muscle relaxation
 - ☞ Suppression of undesirable reflexes (autonomic)
 - ☞ Analgesia

An ideal anesthetic

- Have anxiolytic, sedative & analgesic activity
 - Does not bring about emesis
 - Counter act vagal activity
 - Cause muscle relaxation
 - Possess rapid induction & recovery
 - Chemically stable, non-flammable, non-toxic, easy to administer & inexpensive
- ⊗ No currently available anesthetic agent possesses all these properties at tolerable doses
- ⇒ hence, the need for pre-anesthetic medication arises to produce a balanced anesthesia.

Anesthetic...

Pre-anesthetic medication:

✂ It is the use of drugs prior to anesthesia to make it more safe and pleasant.

- To relieve anxiety & muscle relaxation– benzodiazepines
- To prevent nausea and vomiting–antiemetics
- To provide analgesia – opioids
- To prevent bradycardia and secretion – atropine
- To prevent allergic reactions – antihistamines
- To facilitate intubation and relaxation-NM blocking agents

⇒Balanced anesthesia=pre anesthetic + IV anesthetic (for induction)+ inhalational(for maintenance)

■anesthetic

- general anesthesia is normally induced with an intravenous anesthetic like ***thiopental***, which produces unconsciousness within 25 seconds after injection
- For children, without intravenous access, non-pungent agents, such as ***halothane*** or ***sevoflurane***, are used to induce general anesthesia.
- Opioids, such as ***fentanyl***, are often used for pain along with inhalation agents, because the latter are not good analgesics.

General anesthetics can be classified as:

❖ Inhalational

■ Gas

- ✓ Nitrous oxide

■ Volatile liquids

- ✓ Ether
- ✓ Halothane
- ✓ Enflurane
- ✓ Isoflurane
- ✓ Desflurane
- ✓ Sevofluran

❖ Intravenous

➤ Inducing agents

- ✓ Thiopentone sod.
- ✓ Methohexitone sod.
- ✓ Propofol,
- ✓ Etomidate

➤ Slower acting drugs

- Benzodiazepines
 - ✓ Diazepam
 - ✓ Lorazepam
 - ✓ Midazolam
- Dissociative anaesthesia
 - ✓ Ketamine

- Opioid analgesia---Fentanyl

❖ Nitrous oxide

- ✓ Nitrous oxide is a good analgesic;
- ✓ Onset of action is quick and smooth , recovery is rapid:
- ✓ Post-anaesthetic nausea is not marked.
- ✓ No metabolism,
- ✓ Minimal CV effects,
 - little effect on respiration, heart and BP: breathing and circulation are better maintained
- ✓ Diffusional hypoxia
- ✓ It is a **poor muscle relaxant**; neuromuscular blockers are often required
- ✓ cheap and very commonly

- **Halothane**

- ✓ volatile liquid & noninflammable.
- ✓ intermediate induction is reasonably quick and pleasant
- ✓ It is **not a good analgesic or muscle relaxant**
- ✓ causes direct depression of myocardial contractility---
Ca²⁺ concentration.
- ✓ Moreover, sympathetic activity fails due to cardiac effect
↓
 - BP starts falling ---Heart rate is reduced
- depression of respiration; breathing is shallow and rapid
- It inhibits intestinal and uterine contractions.
 - ✓ prolong delivery and increase postpartal blood loss.
- Used in obstetrics to induce uterine relaxation
- In children for inhalation induction (unlike adults, not hepatotoxic)

Halothane...

Advantages of halothane:

- pleasant smell, thus, is accepted by many patients
- Very potent anesthetic (MAC= 0.75%)
- Non-irritant
- Bloodless field for plastic surgery because of the fall in blood pressure
- Solubility in blood is intermediate $\Rightarrow$ reasonable induction & recovery
- No post operative nausea & vomiting
- No bronchospasm, no coughing, hence suitable for tracheal intubation.

Disadvantages of halothane:

- Little analgesic activity
- Little muscle relaxation
- Fall in blood pressure because of myocardial depression
- Expensive

- **Minimum Alveolar Concentration(MAC)**
- Refers the minimum concentration of the drug in the alveolar air that will produce immobility in 50% of patients exposed to a painful stimuli
- is index of inhalation anesthetic potency
- Low MAC means the drug has high potency and is needed in small amount-----more lipid soluble
 - ✓ *Eg. halothane, is more potent than nitrous oxide.*

Anaesthetic	Inflamma- -bility	Irritancy	Induction	Musclerelax ation
Ether	Infl. +Explo.	Pungent	Slow	v. good
Halothane	Noninfl.	Pleasant	Interm.	Fair
Enflurane	Noninfl.	Pleasant	Interm.	Good
Isoflurane	Noninfl.	Not pleasant	Fast	Good
Desflurane	Noninfl.	Unpleasant	Fast	Good
Sevoflurane	Noninfl.	Pleasant	Fast	Good
Nitrous oxide	Noninfl.	Pleasant	Fast	Poor

Side effects	Halothane	Isoflurane	Desflurane	Sevoflurane
Arrhythmias	↑	-	-	-
Sensitivity to catecholamines	↑	-	-	-
Cardiac output	Decreased	Decreased to a lesser extent than halothane	Decreased to a lesser extent than halothane	Decreased to a lesser extent than halothane
Blood pressure	Dose dependent decreased	Dose dependent decreased	Dose dependent decreased	Dose dependent decreased
Respiratory reflexes	Inhibited	Initial stimulation	Initial stimulation	Inhibited
Hepatic toxicity	Some risk	Low risk	Low risk	Low risk
Renal toxicity	Low risk	Low risk	Low risk	Some risk

■ Intravenous anesthetics

- are generally employed to induce anesthesia, to provide supplemental anesthesia, or to permit anesthesia for short operative procedures

- These include

- **Short acting barbiturates**

 - Eg Sodium salt of thiopental

- **Bezodiazepines:-** to produce unconsciousness and amnesia, pre operative sedation, induction of anesthesia eg diazepam, midazolam

- **Propofol:** - sedative-hypnotic, induction of anesthesia, antiemetic

- **Others** – ketamine,, etomidate, opioids (eg fentanyl)

- **Thiopentone sod.**

- ✓ high lipid solubility enters brain----- brain gets large
- ✓ less vascular tissues (muscle, fat) it back diffuses from the brain:
- ✓ **poor analgesic**.so, Painful procedures ----opioid or N2O given
- ✓ It is a **weak muscle relaxant**
- ✓ BP falls immediately after injection due to vasodilatation,
- ✓ but recovers rapidly.
- ✓ It does not sensitize the heart to Adr,
- ✓ It can be employed as the **sole anaesthetic** for short operations that are not painful.

- **Adverse effects**

- Laryngospasm
 - ✓ prevented by atropine premedication and administration of succinylcholine immediately after thiopentone.
 - ✓ Succinylcholine and thiopentone not be mixed in the same syringe.
- Shivering and delirium may occur during recovery.
- Pain likely to induce restlessness----- *adequate analgesia*
- nausea and vomiting are **uncommon**.
- It can precipitate acute intermittent **porphyria** in susceptible individuals-contraindicated.

- **Propofol**

- ✓ propofol has superseded thiopentone for induction & maintenance.
- ✓ much shorter than that of thiopentone
- ✓ It lacks airway irritancy
- ✓ nausea and vomiting is low;
- ✓ Fall in BP with less marked cardiac depression
- ✓ respiratory depression more marked than thiopentone.
- ✓ Pain during injection(emulsion form)-----***minimized with lidocaine.***
- ✓ drug of choice for sedating intubated patients
- ✓ it is not use in children;
 - prolonged sedation with higher doses -----metabolic effects and heart failure even in adults
- Potentiating the chloride current through the GABA A receptor

Actions

✓ **CNS** effects- depresses, accompanied by excitatory phenomena, such as muscle twitching, spontaneous movement, yawning, and hiccups

✓ **Cardiovascular Effects**-decreases BP without depressing the myocardium (arterial & veno dilation)

✓ **Respiratory Effects**

- is a potent respiratory depressant and generally produces apnea after an induction dose

✗ **Fospropofol**

✓ is a water soluble prodrug of propofol

✓ The prodrug is rapidly metabolized to propofol,

• **SLOWER ACTING DRUGS**

❖ **Benzodiazepines:**

- ✓ In addition to preanaesthetic medication, used for inducing, maintaining, supplementing, sedation.
- ✓ Recovery is further **delayed**
- ✓ poor analgesics:
- ✓ Require neuromuscular blocking drugs for muscle relaxation
- ✓ not provoke postoperative nausea
- ☑ Compared with the intravenous barbiturates, benzodiazepines produce a **slower onset** of central nervous system depressant effects.
- ☑ It lacks the tendency to cause **respiratory** and **cardiovascular** depression.

Benzodiazepines....

- ☑ Midazolam is water-soluble and is the BZP of choice for parenterally administration-----**nonirritating to veins**
 - at physiologic pH midazolam becomes lipid-soluble and can cross the BBB to produce its central effects
 - ✓ Has rapid onset & short $t_{1/2}$
 - ✓ 3 times more potent than diazepam.
 - ✓ It is being preferred over diazepam for anaesthetic use:
- ☑ Diazepam and Lorazepam are not water-soluble, and their IV use necessitates non-aqueous vehicles, which cause pain and local irritation.

❖ Ketamine

- ✓ profound analgesia, immobility, amnesia with light sleep
- ✓ Respiration is not depressed; eyes may remain open.
- ✓ Heart rate, cardiac output and BP are elevated due to sympathetic.
- ✓ Children tolerate the drug better
- ✓ used for operations on the head and neck, in asthmatics
- ✓ who do not want to lose consciousness
- ✓ It is good for repeated use;
- ✓ dangerous for hypertensives, in ischaemic heart disease , but is good for hypovolemic patients.

❖ Ketamine

✓ **Highly lipophilic** drug and is rapidly distributed into well-perfused organs (the brain, liver, and kidney).

- The only intravenous anesthetic that possesses anesthetic and analgesic properties, as well as the ability to produce dose-related cardiovascular stimulation.
- blockade **NMDA receptor** subtype

✓ **Adverse Effect**

- postoperative disorientation
- sensory and perceptual illusions
- vivid dreams
- Increase intracranial pressure

Etomidate

- ✓ A hypnotic agent but lacks analgesic activity
- ✓ Poor water solubility, so is formulated in a propylene glycol solution.
- ✓ Rapid but short-acting
- ✓ Only used for patients with coronary artery disease or cardiovascular dysfunction, such as shock.

ADE

- Decrease in plasma cortisol and aldosterone levels

Opioid Analgesics

- ♣ Fentanyl, sufentanil, remifentanyl, alfentanil
- ♣ IV, Epidurally, or Intrathecally
- ♣ Analgesic property $\Rightarrow$ used together with anesthetics
- ♣ High doses OAs + large doses of BZPs for major cardiovascular surgical procedures
- ♣ Fentanyl + droperidol to produce analgesia and amnesia and combined with N₂O provide a state referred to as neuroleptanesthesia

❖ Fentanyl

- ✓ short acting potent opioid analgesic-----***abolished pain*** .
- ✓ Combined with BZDs, replaced inhaled anesthetics for diagnostic, endoscopic, angiographic and other minor procedures
- ✓ Nausea, vomiting and itching during recovery is common
- ✓ Fall in BP is slight and heart is not sensitized to Adr
- ✓ The opioid antagonist naloxone can be used to counteract persisting respiratory depression and mental clouding.
- ✓ adjunct to spinal and nerve block anaesthesia,

Therapeutic Disadvantages

- Must be delivered using a special vaporizer

- Incomplete anesthesia
- No muscle relaxation
- Must be used with other anesthetics for surgical anesthesia

- Reduces hepatic and renal blood flow
- Lowers blood pressure
- Sensitizes myocardium to actions of catecholamines
- Hepatic toxicity
- Arrhythmias

- Potential renal toxicity at low flows

- Poor analgesia
- Causes significant nausea
- Little muscle relaxation
- Laryngospasm

- Poor analgesia

Inhalation anesthetics

Desflurane

Nitrous oxide

Halothane

Isoflurane

Sevoflurane

Intravenous anesthetics

Thiopental

Ketamine

Fentanyl

Propofol

Dexmedetomidine

Therapeutic Advantages

- Good analgesia
- Rapid onset/recovery
- Safe, nonirritating

- Good muscle relaxation
- Rapid recovery
- Stability of cardiac output
- Does not raise intracranial pressure
- No sensitization of heart to epinephrine

- Bronchial smooth muscle relaxation good for patients with asthma
- Rapid onset/recovery
- Not irritating; useful in children

- Rapid onset of action
- Potent anesthesia

- Good analgesia

- Not likely to cause nausea
- Rapid onset
- Lowers intracranial pressure

- No respiratory depression
- Blunts undesirable cardiovascular reflexes

■ *Local anesthetics*

- Produce regional anesthesia/block pain conduction by nerves without generalized depression of the CNS
- **MOA:-**block sodium channel in axonal membranes
 - ☛ Used as sole anesthetics
 - ☛ also in combination regimens with inhaled and intravenous agents.
- Co-administration of alpha adrenoceptor agonists (e.g., epinephrine) will decrease local anesthetic absorption into the systemic circulation, prolonging effects and possibly decreasing toxicity

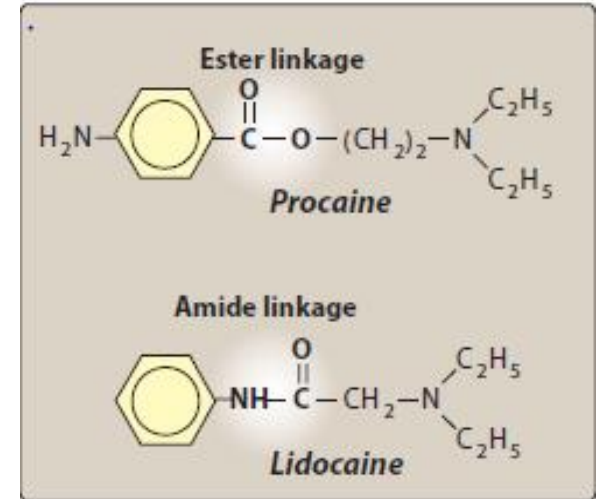
Comparative features of general and local anaesthesia

	<i>General anaesthesia</i>	<i>Local anaesthesia</i>
1. Site of action	CNS	Peripheral nerves
2. Area of body involved	Whole body	Restricted area
3. Consciousness	Lost	Unaltered
4. Care of vital functions	Essential	Usually not needed
5. Physiological trespass	High	Low
6. Poor health patient	Risky	Safer
7. Use in non-cooperative patient	Possible	Not possible
8. Major surgery	Preferred	Cannot be used
9. Minor surgery	Not preferred	Preferred

■ Local anesthetics can be esters or amides

■ Amides

- Lidocaine, Bupivacaine, Levobupivacaine, Ropivacaine, Mepivacaine, Etidocaine, Prilocaine
 - ✓ More likely systemic toxicity
- Slowly metabolized by liver amidases



■ Esters

- Procaine, Chlorprocaine, Tetracaine, Benzocaine, Cocaine (the first local anesthetic that has largely been replaced by synthetic agents)
 - ✓ Less likely systemic toxicity
- Rapidly metabolized by plasma and tissue esterases

- **Lidocaine (Lignocaine)**

- currently the most widely used LA.
- It is a versatile LA, surface application, infiltration, nerve block, epidural, spinal and intravenous regional block anaesthesia
- Injected around a nerve it blocks conduction
 - ✓ within 3 min, whereas procaine may take 15 min; also anaesthesia is more intense and longer lasting.
 - ✓ Vasodilatation occurs in the injected area.
 - ✓ Cross sensitivity with ester LAs is not seen.

- **Actions**

- ✓ cause vasodilation,
 - ✓ leading to rapid diffusion away from the site of action and shorter duration when these drugs are administered alone.
- ✓ By adding the vasoconstrictor *epinephrine*, the
 - ✓ rate of local anesthetic absorption and diffusion is decreased.
 - ✓ minimizes systemic toxicity and increases the duration of action.
- ✓ Hepatic function does not affect the duration of action of local anesthesia,
- ✓ Some local anesthetics have other therapeutic uses (for example, *lidocaine* is an IV antiarrhythmic).

- **Therapeutic use**

- used to prevent and relieve pain from medical procedures, disease, or injury
- may also be used for severe pain that topical anesthetics or analgesics can't relieve

- **Adverse effects**

- CNS depression, cardiovascular depression, allergic reaction (more commonly with ester types), depressed uterine contractility

Classification

① Topical or Surface anesthesia

- Applied to skin or mucous membrane
- Used to treat pain, itching stimuli on conjunctiva, larynx, throat, hemorrhoids, damaged skin

⇒ Benzocaine, cocaine, pramoxine

② Infiltration anesthesia

- Injected directly into or very close to the area to be anesthetized.
- Used during tooth extraction, surgical incision
- Etidocaine, lidocaine, prilocaine & tetracaine

Classification

③ Nerve block anesthesia

- Injected as close as possible to the nerve trunk supplying the specific area to be anesthetized
- Enable minor operations on the limb

⇒ Lidocaine, mepivacaine, bupivacaine

④ Spinal anesthesia

- Injecting the anesthesia in to CSF, usually in the lumbar spine
- The anesthesia blocks sensory impulses at the root of peripheral nerves as they enter the spinal cord.
- Useful for surgery of lower abdomen & legs

⇒ Procaine, tetracaine

• **Precautions and interactions**

- Given IM or SC
 - Take care to avoid IV injection
- ✓ Propranolol may reduce metabolism of lidocaine and other amide LAs by reducing hepatic blood flow.
- ✓ adrenaline containing LA should be avoided for patients with
 - ischaemic heart disease,
 - cardiac arrhythmia,
 - thyrotoxicosis,
 - uncontrolled hypertension, and those receiving β blockers (rise in BP due to unopposed α action) or tricyclic antidepressants (uptake blockade of Adr).

■ **Therapeutic use**

- relieve or prevent pain, especially minor burn pain
- relieve itching and irritation
- anesthetize an area before an injection is given
- numb mucosal surfaces before a tube, such as a urinary catheter, is inserted
- alleviate sore throat or mouth pain when used in a spray or solution

■ **Adverse effects**

- These drugs may cause topical skin reactions eg irritation

<i>Site</i>	<i>Drugs</i>	<i>Form</i>	<i>Purpose</i>
1. Eye	Tetracaine 1-2% Benoxinate 0.4%	ointment, drops drops	tonometry, surgery tonometry
2. Nose, ear	Lidocaine 2-4% Tetracaine 1-2%	drops	painful lesions, polyps
3. Mouth, throat	Benzocaine	lozenges	stomatitis, sore throat
4. Pharynx, larynx, trachea, bronchi	Lidocaine 2-4% Tetracaine 1-2%	spray	tonsillectomy, endotracheal intubation, endoscopies
5. Esophagus, stomach	Oxethazaine 0.2%	suspension	gastritis, esophagitis, heartburn
6. Abraded skin	Tetracaine 1% Benzocaine 1-2% Butamben 1-2%	cream, ointment, dusting powder	ulcers, burns, itching dermatoses
7. Intact skin	Eutectic lidocaine/ prilocaine 5%	cream under occlusion	i.v. cannulation, skin surgery
8. Urethra	Lidocaine 2%	jelly	for dilatation, catheterisation
9. Anal canal, rectum	Lidocaine 4% Dibucaine 1% Benzocaine 5%	ointment, cream, suppository	fissure, painful piles, surgery, proctoscopy

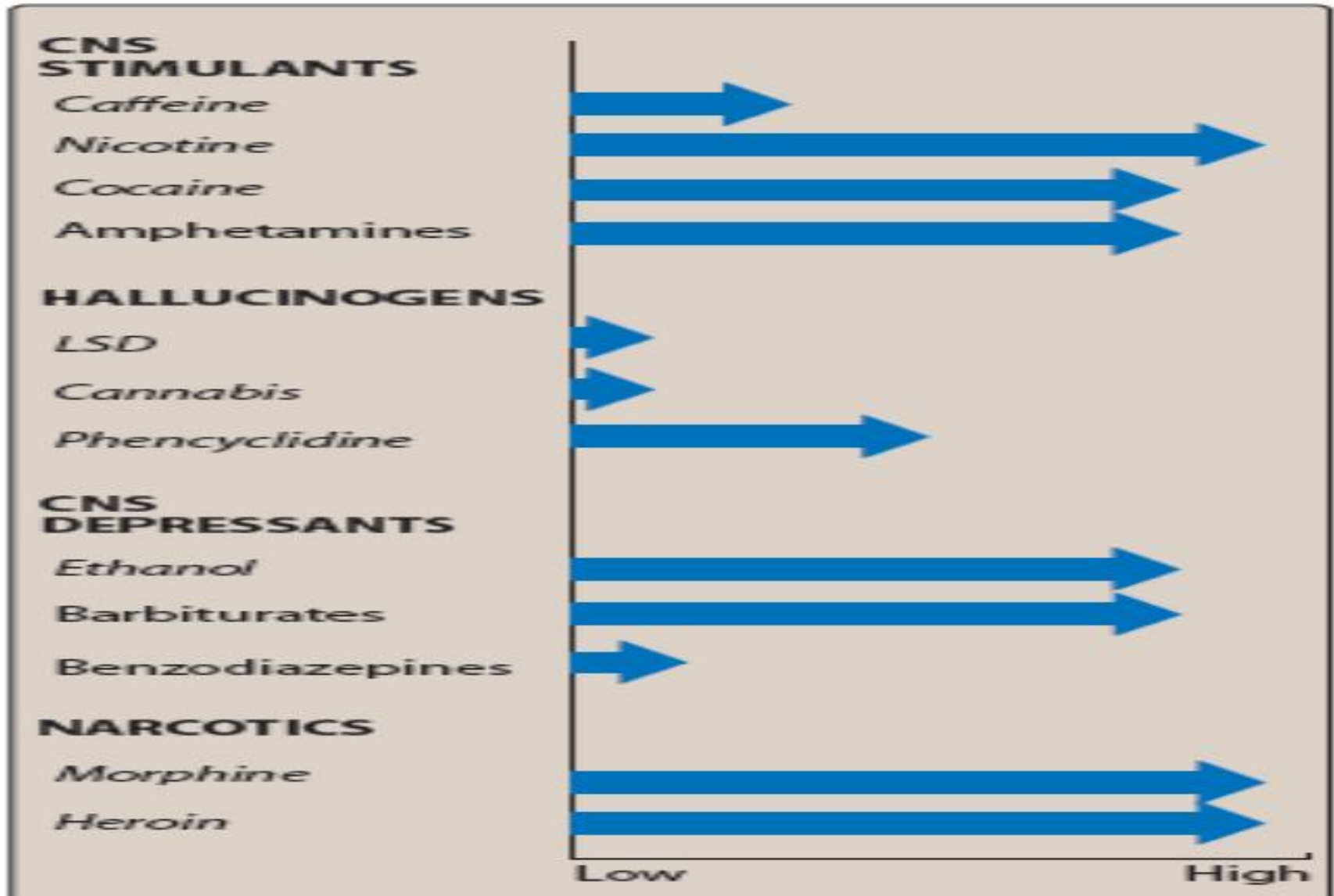
CHARACTERISTIC	ESTERS • Procaine • Chloroprocaine } Tetracaine • Cocaine	AMIDES • Lidocaine • Bupivacaine • Ropivacaine • Mepivacaine • Prilocaine
Metabolism	Rapid by plasma cholinesterase	Slow, hepatic
Systemic toxicity	Less likely	More likely
Allergic reaction	Possible- PABA derivatives form	Very rare
Stability in solution	Breaks down in ampules (heat, sun)	Very stable chemically
Onset of action	Slow as a general rule	Moderate to fast
pK _a 's	Higher than physiologic pH (8.5–8.9)	Close to physiologic pH (7.6–8.1)

DRUG	POTENCY	ONSET	DURATION
Procaine	Low	Rapid	Short
Chloroprocaine	Low	Rapid	Short
Tetracaine	High	Slow	Long (spinal)
Lidocaine	Low	Rapid	Intermediate
Mepivacaine	Low	Moderate	Intermediate
Bupivacaine	High	Slow	Long
Ropivacaine	High	Moderate	Long

❖ CNS stimulants

- two groups of drugs that act primarily to stimulate CNS.
- The first group, the psychomotor stimulants, cause **excitement** and **euphoria**, decrease feelings of fatigue, and increase motor activity.
- The second group, the **hallucinogens**, or **psychotomimetic** drugs, produce profound changes in thought patterns and mood, with little effect on the brainstem and spinal cord.

Relative potential for physical dependence on commonly abused substances



Thank you!